

Materials Challenges for High Performance Magnetocaloric Refrigeration Devices

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Magnetocaloric materials with a Curie temperature near room temperature have attracted significant interest for some time due to their possible application for high-efficiency refrigeration devices. This review focuses on a number of key issues of relevance for the characterization, performance and implementation of such materials in actual devices. The phenomenology and fundamental thermodynamics of magnetocaloric materials is discussed, as well as the hysteresis behavior often found in first-order materials. A number of theoretical and experimental approaches and their implications are reviewed. The question of how to evaluate the suitability of a given material for use in a magnetocaloric device is covered in some detail, including a critical assessment of a number of common performance metrics. Of particular interest is which non-magnetocaloric properties need to be considered in this connection. An overview of several important materials classes is given before considering the performance of materials in actual devices. Finally, an outlook on further developments is presented.

substance upon adiabatic magnetization or demagnetization. This can be qualitatively explained as an interplay between spin and lattice degrees of freedom: upon magnetization the spins order, causing the magnetic entropy to decrease. Under adiabatic conditions this entropy decrease must be balanced by an increase in entropy of the lattice, i.e. an increase in temperature. Alternatively, under isothermal conditions the substance will experience a change in entropy corresponding to a flux of heat to its surroundings. In principle all magnetic substances exhibit the magnetocaloric effect. However, it is most pronounced near magnetic phase transitions where even a small external field can change the spin state significantly. Thus, except at very low (sub-kelvin) temperatures where paramagnets can be utilized, the relevant magnetocaloric materials for cooling purposes

1. Introduction

Magnetocaloric materials with a Curie temperature near room temperature have attracted significant interest for some time due to their possible application for high-efficiency, environmentally friendly heat pumps and refrigerators.^[1,2] The potential for high efficiency is due to the reversible nature of the magnetocaloric effect, while the avoidance of the volatile refrigeration gases used in conventional vapor compression devices, and the possibility of recycling materials and magnets at end-of-life, makes the technology attractive from an environmental point of view. Quite aside from these technological considerations, the magnetocaloric effect has considerable scientific interest in itself as a manifestation of the intricate interactions between the magnetic degrees of freedom and the underlying lattice.

In its simplest form the magnetocaloric effect consists in a reversible temperature change experienced by a magnetic

are magnetic materials with a transition temperature near the temperature region of interest. In the present review the focus will be on materials for applications near room temperature.

The magnetocaloric effect is fully characterized by the two quantities mentioned above, the adiabatic temperature change, ΔT_{ad} , and the isothermal entropy change per unit mass, ΔS , mapped out as functions of field and temperature. The adiabatic temperature change is defined as the difference in temperature when a sample initially is subject to an external magnetic field H_i which is then changed under adiabatic conditions to a final field H_f . The isothermal entropy change is similarly defined as the change in entropy of a sample under isothermal conditions when the external field is changed from H_i to H_f (see Figure 1). It is not possible to calculate one of these quantities from the other. However, both may be inferred from a complete measurement of the field dependent specific heat capacity in constant field, $C_H(T, H)$.^[3] In the normal magnetocaloric effect, ΔT_{ad} will be positive and ΔS negative when the final field is larger than the initial field. It should also be mentioned that the usefulness of a magnetocaloric material for refrigeration purposes cannot be evaluated solely by a knowledge of ΔT_{ad} and ΔS since the heat capacity and a number of non-magnetocaloric properties play a crucial role for the performance in a refrigeration cycle.^[4] This will be a key point in the following.

For most applications the working field has to be generated by permanent magnets, limiting the available maximum fields to 1–2 T (it is usual to give the magnitude of the field in terms

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of $\mu_0 H$, measured in tesla).^[5] For such fields the magnitude of the magnetocaloric effect is typically a few degrees for ΔT_{ad} and 10–20 J/kg·K for ΔS (see **Figure 2**). For example, gadolinium, which is often considered as the benchmark magnetocaloric material for room-temperature applications (having a Curie temperature of $\sim 20^\circ\text{C}$), has a maximum adiabatic temperature change of $\sim 5.8\text{ K}$ for a field change going from 0 to 2 T and a corresponding maximum magnitude of the isothermal entropy change of $\sim 10.4\text{ J/kg}\cdot\text{K}$.^[6] (The precise values depend somewhat on the impurity content.) This means that the materials cannot be used directly for refrigeration (as in paramagnetic demagnetization) but must be part of a regenerative cycle to make the device temperature span exceed the adiabatic temperature change. This is usually achieved by having a magnetic refrigeration device consist of a porous matrix of magnetocaloric material through which a heat transfer fluid passes back and forth. The matrix will then act as a thermal regenerator. In this connection, the significance of the two fundamental magnetocaloric quantities is that ΔS is a measure of how much heat can be moved (at a given temperature) by magnetic means, while ΔT_{ad} is a measure of how high a temperature difference is available to achieve the transfer of the heat to and from the heat transfer fluid. It is not possible to evaluate the performance of a magnetocaloric material based on only one of these quantities. If for instance a material has a high ΔS but a low ΔT_{ad} the heat transfer from matrix to fluid will be slow, limiting the operation frequency and giving rise to significant internal losses.

Several useful materials reviews already exist in the literature.^[7–13] In the present review we will take a complementary approach where the focus will be very much on how to characterize and compare materials in a meaningful fashion and to what extent it is possible to relate measured magnetocaloric properties to the performance of a material in a device. This includes a discussion of performance metrics such as refrigerant capacity which are often quoted in the literature, and a critical examination of the different characterization methods used. The review is organized as follows: after a brief historical introduction we survey the different types of magnetocaloric materials and their phenomenology before considering the fundamental thermodynamics of the magnetocaloric effect, including the applicability of the so-called Maxwell relation connecting the magnetization properties to the isothermal entropy change. Then the phenomena associated with first-order materials are reviewed, in particular the hysteresis behavior often found in such materials. This is followed by an overview of some common theoretical approaches, both for first-order and second-order materials, and the question of the scaling properties of the magnetocaloric effect and fundamental limits to its size. Turning to experiment, a critical discussion of the most common characterization techniques is given, including both magnetization measurements, calorimetric methods and direct evaluation of the magnetocaloric effect. We then consider in some detail how to evaluate the suitability of a given material for use in a magnetocaloric device. This important point is not always addressed with sufficient care; of particular interest for actual devices is the question of which non-magnetocaloric properties need to be considered in this connection. Next, the question of using series of materials in a graded configuration is addressed. Then an overview of several important materials classes is given before considering



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materials performance in actual devices. Finally, an outlook on further developments is presented.

2. Brief historical survey

Conventionally, but inaccurately, the discovery of the magnetocaloric effect is ascribed to Warburg who in 1881 discussed the increase in temperature in iron wires submitted to a cycle of magnetization and demagnetization.^[14] Several remarks may be made in that connection: 1) Warburg's field source was not very strong, only up to 100 times the horizontal component of the Earth's magnetic field in Freiburg, i.e. perhaps 50 gauss = 0.05 T. 2) His experiments were done at ambient temperature. Given that the Curie temperature of iron is 770°C , the adiabatic temperature change in iron is by usual methods unobservably

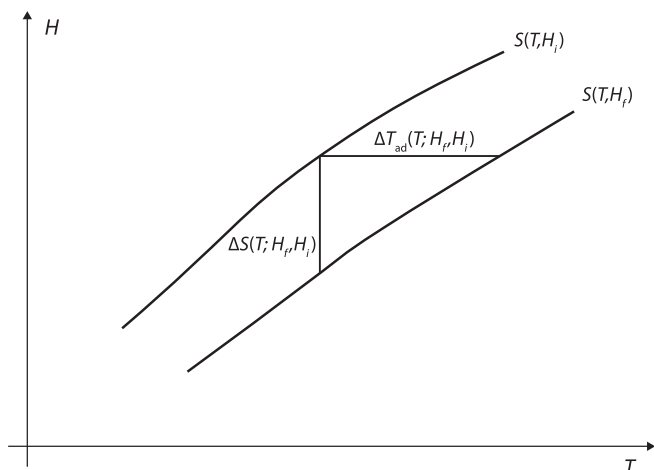


Figure 1. The basic magnetocaloric quantities ΔT_{ad} and ΔS defined in terms of the entropy curves $S(T, H_i)$ and $S(T, H_f)$.

small at room temperature even in sizable fields.^[15,16] Only by special methods capable of resolving temperature differences on the order of 10^{-6} K is it possible to observe the magnetocaloric effect in this range.^[17] 3) Warburg did not measure the temperature, neither directly nor indirectly. Instead he calculated the work performed on the iron wire as the area of the hysteresis loop described upon magnetization and demagnetization. This he translated into a corresponding temperature change of a quantity of water. 4) Given that the magnetocaloric effect in iron is reversible and that Warburg's cycle included both magnetization and demagnetization, the corresponding total magnetocaloric temperature change is zero. From these points it should be clear that Warburg did not discover the magnetocaloric effect — which indeed he did not claim to have done. Rather, he proposed what is sometimes known as Warburg's law: that the work done on a magnetic body submitted to a hysteresis loop is equal to (μ_0 times) the area enclosed by the loop in an $M-H$ -diagram.^[18–20]

In the late 19th century both Edison and Tesla took out patents on “pyromagnetic generators”.^[21,22] These took advantage of the change in magnetic properties of iron when cyclically heated above and cooled below its Curie temperature by an external source of heat. In this way, electricity could be generated from

heat. Although unrelated to the magnetocaloric effect, these patents showed an increasing awareness of what may loosely be called magnetothermal effects.

In 1905 Langevin showed on the basis of the rudimentary electron theory as it existed at that time, that a paramagnet should exhibit a reversible temperature change upon magnetization or demagnetization. He estimated that the temperature change in oxygen at ordinary temperatures would amount to 1/1000 of a degree in a field of 1 T, making the question of observing this effect largely academic.^[23]

Then, in 1918, Weiss and Piccard discovered that in nickel a reversible change in temperature can be observed in the vicinity of the Curie temperature (354 °C);^[24] they found a temperature change of 0.7 K in a field of 1.5 T. Thus, they correctly identified the two distinguishing features of the ordinary magnetocaloric effect: that it is reversible and that it is largest at temperatures near the Curie temperature. This marks the real discovery of the magnetocaloric effect. The authors also gave the first thermodynamic explanation of the effect which was expanded in the following years by Weiss and coworkers^[25] as well as Debye.^[26] In the mid-1920s Debye and Giaucque independently of each other realized that the magnetocaloric effect can be used for refrigeration at very low temperatures.^[26,27] This could be done by adiabatic demagnetization of paramagnetic salts, specifically gadolinium sulfate, $Gd_2(SO_4)_3 \cdot 8H_2O$. Due to the relative smallness of the effect in paramagnets this limited the application to very low temperatures where the specific heat of the substance to be cooled is sufficiently small. Adiabatic demagnetization as a means of attaining very low temperatures was demonstrated in the early 1930s and was soon developed into a standard low temperature laboratory technique.^[28]

Until the discovery of the ferromagnetism of gadolinium in 1935 by Urbain, Weiss and Trombe^[29] no ferromagnetic substance with a Curie point near room temperature was known. Curiously, in view of Weiss being co-discoverer, the magnetocaloric effect of gadolinium seems not to have been studied until more than 20 years later when it was investigated by a group at West Virginia University. As far as can be ascertained their results were only made available in two unpublished Master's theses (unavailable to the present authors).^[30,31]

The concept of near-room-temperature magnetic refrigeration using the magnetocaloric effect can be dated from the remarkable paper by Brown who in 1976 not only suggested that Gd

would be suited for such an application but also demonstrated a laboratory device which utilized 157 g (20 cm³) of Gd arranged as 1 mm thick plates separated by screen wire.^[32] A liquid column consisting of 400 cm³ of a water/ethanol mixture functioned as a thermal regenerator, allowing a no-load temperature difference of 47 K (−1 °C to 46 °C) to be produced using a 7 T superconducting magnet. This prompted the development of further devices, e.g. by Steyert in 1978.^[33] In 1982 Barclay and Steyert patented the active magnetic regenerator (AMR) in which the magnetocaloric material does double duty as a thermal regenerator.^[34] This simplifies the design and still allows temperature spans well

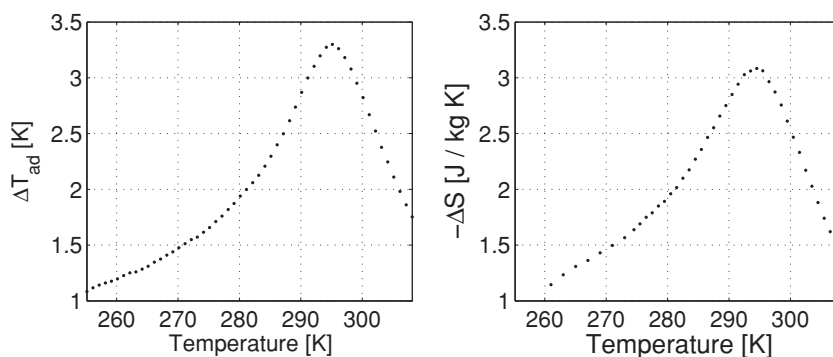


Figure 2. The magnetocaloric effect in commercial grade gadolinium in a field of 1 T, as experimentally determined.

in excess of the adiabatic temperature change to be achieved. Almost all subsequent devices have used some variation of the AMR principle which is believed to be the most efficient design as regards thermodynamical efficiency,^[35] although no rigorous proof of this has been given in the literature. In 1997 a proof-of-principle device capable of delivering a maximum cooling power of 600 watt was developed in a collaboration between Ames Laboratory and Astronautics Corporation in the US.^[36] This used a superconducting magnet but the same team developed a device based on permanent magnets which became operational in 2001.^[13] On the materials side a breakthrough occurred in 1997 with the discovery of the so-called giant magnetocaloric effect in $\text{Gd}_5\text{Si}_2\text{Ge}_2$ by Pecharsky and Gschneidner.^[37] They observed a magnetocaloric effect exceeding that of all hitherto known magnetic materials, including gadolinium, around a first-order transition at 276 K between two different ferromagnetic phases. This spurred a dramatic increase in the research effort worldwide on both magnetocaloric materials and device design. Several new materials classes with promising characteristics have been discovered, and a recent device review counted more than 40 working lab devices reported in the literature.^[38] At least two additional companies, Camfridge Ltd. in the UK and Cooltech Applications in France, have stated that they will present prototypes in the near future.

3. Classification and phenomenology of magnetocaloric materials

Magnetocaloric materials may be classified according to their physical properties in a number of ways. A fundamental distinction is between the *ordinary* and *inverse* magnetocaloric effect. In the ordinary magnetocaloric effect the material heats up upon magnetization and cools down upon demagnetization; the corresponding isothermal entropy change is then negative (positive) upon magnetization (demagnetization). In the inverse magnetocaloric effect the material cools down upon magnetization. This cannot be explained in the simple picture used above but can occur if the temperature derivative of the magnetization is positive, as can be the case in antiferromagnetic and ferrimagnetic systems.^[39]

The type of magnetic order offers a further way of classifying the materials. *Paramagnets* exhibit the magnetocaloric effect but since the associated relative entropy change is minute except at very low temperatures, they will not be considered further here. *Ferromagnets* may be classified according to the order of the transition. *Second-order materials* exhibit a classical second-order ferromagnetic to paramagnetic transition at some temperature T_c characterized by the fact that the intrinsic spontaneous magnetization goes continuously to zero when the temperature approaches T_c from below. *First-order materials* have a transition in which the magnetization changes discontinuously at some temperature and there will be a latent heat associated with the phase transition. In real systems, impurities and spatial variations in the stoichiometry may blur the first-order/second-order distinction by spreading out the transition and making the first-order transition continuous. Thus, what constitutes reliable experimental evidence of a first-order transition is still a matter of some debate in the literature.^[40]

Within first-order materials a wide variety of different transition types can arise. *Metamagnetic* transitions, i.e. transitions between two different magnetic structures, can be, e.g., ferromagnetic to ferromagnetic, antiferromagnetic to ferromagnetic, and ferrimagnetic to ferromagnetic. In many cases they are field induced and can occur together with structural phase transitions, giving rise to rich phase diagrams. Transitions from a ferromagnetic state to a non-magnetic state with a different structure can be of the *martensitic* type where there is a diffusionless transition between the two structures; many examples may be found among the Heusler alloys. Such alloys may also exhibit the inverse magnetocaloric effect if they have a martensitic transformation from a low-temperature paramagnetic phase to a high-temperature ferromagnetic phase.^[41] A novel type of magnetism, *mixed magnetism*, has been identified by Dung et al.^[42] This consists in the coexistence of alternating atomic layers of weak and strong magnetism which the authors argue can give rise to a significant magnetocaloric effect.

Although *hysteresis* should not be taken as the defining characteristic of a first-order transition,^[43] it can arise in first-order transitions due to the specifics of the microstructural state of the material. The hysteresis may be in both temperature and/or magnetization. This will be discussed further below.

Disregarding hysteresis for the present, it is to some extent possible to identify generic features associated with second-order and first-order materials, respectively. For second-order materials, the maximum magnitude of the isothermal entropy change and the maximum adiabatic temperature change increase with increasing field as the field to some power n which is smaller than one; it is usually close to $\frac{2}{3}$ which is also a prediction of the mean field theory (see below, Section 6.2). For first-order materials, the increase with field is also sublinear. Due to this sublinear dependence on the field, the observed effect is a function of both the initial field and the final field, not just their difference. This is illustrated in **Figure 3**. An elementary consequence is that to maximize the magnetocaloric effect it is more important to ensure that H_i is very small than to increase H_f marginally; this

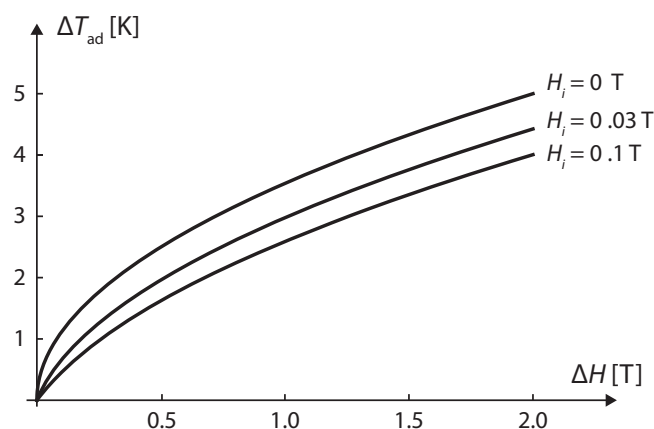


Figure 3. The adiabatic temperature change ΔT_{ad} of a generic magnetocaloric material as a function of $\Delta H = H_f - H_i$ for different $H_i = 0, 0.03, 0.1$ T. An exponent of $\alpha = \frac{2}{3}$ is assumed for the field dependence. It is seen that one can get the same performance at a significantly lower ΔH if H_i can be reduced.

observation has a direct impact on the design constraints on permanent magnet arrays for refrigeration devices.^[44] Another consequence is that it is not very useful to report, e.g., the adiabatic temperature change per tesla of applied field. Nevertheless, this is often done in the literature.

For second-order materials the zero-field specific heat curve is usually quite sharply peaked (unless broadened by impurities) as a function of temperature; as the field increases the peak is lowered and broadened, while the peak temperature shifts (weakly) towards higher temperatures. The corresponding curves for the entropy change and adiabatic temperature change increase in height and width, while the peak positions also changes weakly. Even for quite high fields the peak height of the curves continue to increase.^[6,45]

For first-order materials, the peak in the specific heat may move significantly with field while the width of the peak hardly increases. This can intuitively be understood in terms of a first-order transition between a low temperature ferromagnetic state and a high temperature paramagnetic state: the effect of increased field will be to stabilize the low temperature state (i.e. shifting the transition temperature up) but the phase transition will still be first order (i.e., still sharp). From this behavior of the specific heat follows the often observed characteristics of the isothermal entropy change for a first-order material: the peak height of $-\Delta S$ is not very sensitive to the size of the external field (for large fields) while the width of the peak scales nearly linearly with magnetic field.^[46,47]

In Figure 4 generic curves for second-order and first-order materials are shown, based on a minimal phenomenological model of C_H where the peak either moves weakly while broadening and decreasing (second-order materials) or simply moves as a function of field without changing shape (first-order materials). It should be emphasized that actual materials exhibit a much wider range of magnetocaloric response.

4. The thermodynamics of the magnetocaloric effect

The standard equilibrium thermodynamics of magnetic solids allows one to derive a number of useful relations between the magnetocaloric properties and the magnetic state of the material. In the following we consider only isobaric processes, i.e. we will not consider the pressure dependence of any physical properties. The usual starting point when considering the magnetocaloric effect is the following expression for the entropy

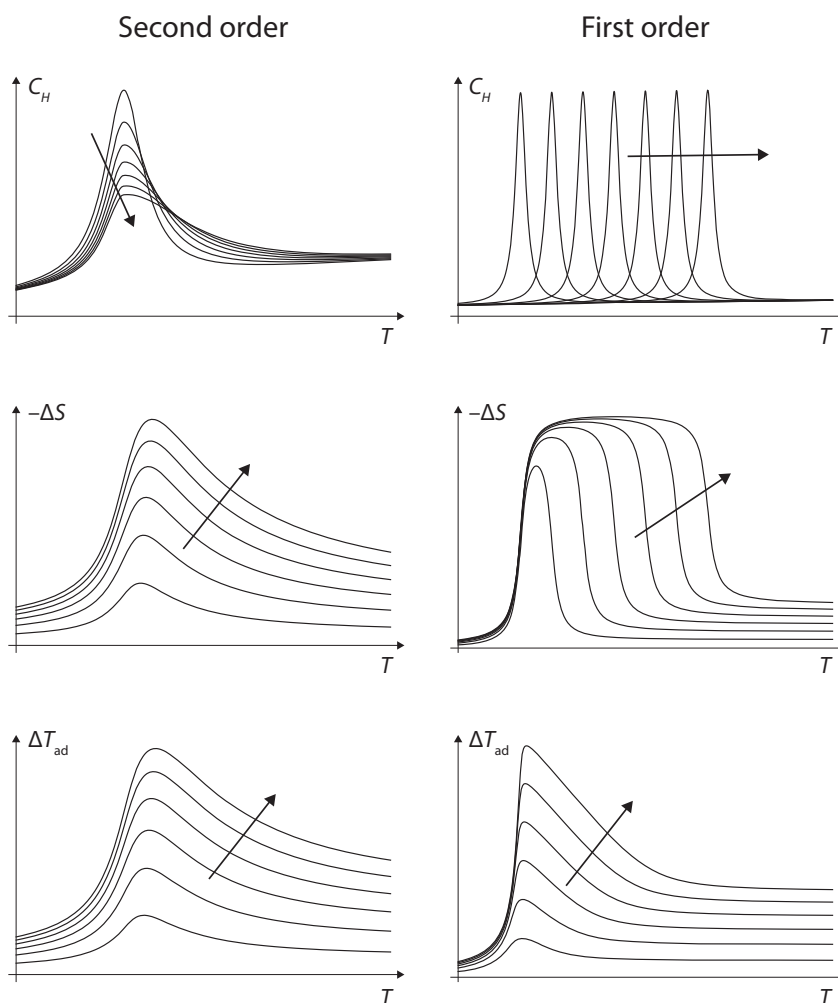


Figure 4. Heat capacity $C_H(T, H)$, isothermal entropy change $\Delta S(T; H, 0)$ and adiabatic temperature change $\Delta T_{ad}(T; H, 0)$ for generic second-order and first-order materials in different external fields in the vicinity of their phase transition; ΔS and ΔT_{ad} have been calculated from the assumed form of C_H using the results derived in Section 4. The arrows indicate the direction of increasing H .

$S(T, H)$ considered as a function of the two independent variables T and H :^[1,48,49]

$$dS = \left(\frac{\partial S}{\partial T} \right)_H dT + \left(\frac{\partial S}{\partial H} \right)_T dH. \quad (1)$$

To get an expression for the adiabatic temperature change ΔT_{ad} , consider the above equation along an isentropic curve γ (i.e. a curve along which $S = \text{const}$) in the (T, H) -space, starting in (T_i, H_i) and ending in (T_f, H_f) . Evidently $\Delta T_{ad} = T_f - T_i$. It is important to note that the adiabatic temperature change is a function of three quantities: the starting temperature, T_i , the starting field H_i , and the final field H_f : $\Delta T_{ad} = \Delta T_{ad}(T_i, H_f, H_i)$. As noted above, ΔT_{ad} is not just a function of the difference in field $\Delta H = H_f - H_i$. Now, along an isentrope Equation 1 becomes

$$0 = \left(\frac{\partial S}{\partial T} \right)_H dT + \left(\frac{\partial S}{\partial H} \right)_T dH. \quad (2)$$

Using the familiar fact that

$$\left(\frac{\partial S}{\partial T}\right)_H = \frac{C_H(T, H)}{T}, \quad (3)$$

where $C_H(T, H)$ is the isofield specific heat capacity, Equation 2 can be rewritten as

$$dT = -\frac{T}{C_H(T, H)} \left(\frac{\partial S}{\partial H}\right)_T dH. \quad (4)$$

It is usual in the literature to say that the above equation is 'integrated' to get

$$\begin{aligned} \Delta T_{ad}(T_i; H_f, H_i) &\equiv T_f - T_i \\ &= - \int_{H_i}^{H_f} \frac{T}{C_H(T, H)} \left(\frac{\partial S}{\partial H}\right)_T dH. \end{aligned} \quad (5)$$

This, however, is somewhat misleading, since the right hand side of Equation 4 obviously is not an exact differential. Thus, the integral is not independent of the integration path — indeed, it must be evaluated along the relevant isentrope γ . A more precise way of writing Equation 5 is therefore by indicating explicitly that the integration is to be carried out along γ .

$$\Delta T_{ad}(T_i; H_f, H_i) = - \int_{\gamma} \frac{T}{C_H(T, H)} \left(\frac{\partial S}{\partial H}\right)_T dH. \quad (6)$$

However, along the isentrope T varies with H (this is the very variation needed to get the adiabatic temperature change), making Equation 6 of limited value for direct evaluation. In practice ΔT_{ad} is evaluated by going back to the differential form in Equation 4 and numerically integrating in sufficiently small increments. It should be mentioned that this is mostly relevant for modeling purposes where closed forms for C_H and S are available, since the heat capacity is rarely sufficiently well-known experimentally as a function of magnetic field to permit the numerical integration to be carried out with adequate precision.

An alternative expression for the adiabatic temperature change in terms of an ordinary integral (i.e. not a line integral) over the magnetic field may be derived from the functional relationship

$$S(T_i, H_i) = S(T_i + \Delta T_{ad}(T_i; H_f, H_i), H_f), \quad (7)$$

which follows directly from the definition of ΔT_{ad} . Differentiating this equation with respect to the initial field gives

$$\begin{aligned} \frac{\partial S}{\partial H_i}(T_i, H_i) &= \frac{\partial S}{\partial T}(T_i + \Delta T_{ad}(T_i; H_f, H_i), H_f) \\ &\quad \cdot \frac{\partial \Delta T_{ad}(T_i; H_f, H_i)}{\partial H_i}. \end{aligned} \quad (8)$$

where care has been taken to indicate which arguments the derivatives of the entropy are to be evaluated in. Rearranging with the use of Equation 3 and integrating over field, the following functional relation for ΔT_{ad} is obtained

$$\begin{aligned} \Delta T_{ad}(T_i; H_f, H_i) &= - \int_{H_i}^{H_f} \frac{T_i + \Delta T_{ad}(T_i; H_f, H)}{C_H(T_i + \Delta T_{ad}(T_i; H_f, H), H)} \\ &\quad \times \frac{\partial S(T_i, H)}{\partial H} dH. \end{aligned} \quad (9)$$

To lowest order in $\Delta T_{ad}/T_i$ the useful approximate relation

$$\Delta T_{ad}(T_i; H_f, H_i) \simeq -T_i \int_{H_i}^{H_f} \frac{1}{C_H(T_i, H)} \frac{\partial S(T_i, H)}{\partial H} dH \quad (10)$$

is obtained. For temperatures around room temperature $\Delta T_{ad}/T_i$ is at most 1-2% in realistically obtainable fields. Furthermore for materials in which the phase transition is of second order, the temperature scale over which $C_H(T, H)$ varies appreciably is generally much larger than ΔT_{ad} for all but the lowest fields (where the peak in C_H is sharpest). Thus, in this case Equation 10 will be accurate to within a few percent. If the field dependence of C_H is entirely ignored (which is a questionable assumption, even for second-order materials)^[50] Equation 10 can be simplified even further:

$$\Delta T_{ad}(T_i; H_f, H_i) \simeq -\frac{T_i}{C_H(T_i)} \Delta S(T_i; H_f, H_i), \quad (11)$$

relating the adiabatic temperature change directly to the isothermal entropy change ΔS . Although a rather crude approximation, this form is often encountered in the literature.

However, if the zero-field specific heat in Equation 11 is replaced with the specific heat evaluated at a non-zero field, which for convenience can be taken to be H_f (assumed to be the higher field), the approximation is improved. Thus,

$$\Delta T_{ad}(T_i; H_f, H_i) \simeq -\frac{T_i}{C_H(T_i, H_f)} \Delta S(T_i; H_f, H_i), \quad (12)$$

allows the adiabatic temperature change for second-order materials to be calculated with reasonable accuracy from a knowledge of the isothermal entropy change and the specific heat measured at a *single* non-zero field (or, alternatively, the specific heat averaged over a suitable field interval). Equation 12 was derived by M. Foldeaki et al.^[51,52] using a somewhat different approach, subsequently criticized by Pecharsky and Gschneidner.^[50] The present approach shows that Equation 12 can indeed be defended as a systematic approximation to the full expression for ΔT_{ad} , Equation 9. In **Figure 5** a comparison between the exact calculation based on Equation 4 and the three approximations in Equation 10, 11 and 12 is shown within a mean field model for gadolinium. It is apparent that Equation 11 is a poor approximation, but also that even Equation 12 performs not too badly, slightly overestimating ΔT_{ad} below T_c and slightly underestimating it above.

For first-order materials the use of Equation 12 is usually unjustified, as can also be seen from **Figure 4** where it is evident that the shape of the ΔS and ΔT_{ad} curves can differ significantly.

Finally we note that from Equation 3 it follows that the isothermal entropy change can be calculated from the heat capacity as

$$\Delta S(T; H_f, H_i) = \int_0^T \frac{C_H(T, H_f)}{T} dT - \int_0^T \frac{C_H(T, H_i)}{T} dT. \quad (13)$$

And according to Equation 6, ΔT_{ad} may be calculated from $S = \int_0^T C_H dT/T$ and C_H . (The zero temperature entropy is assumed to be 0; it is in any case field-independent and therefore plays no role here.) Alternatively, it may be determined

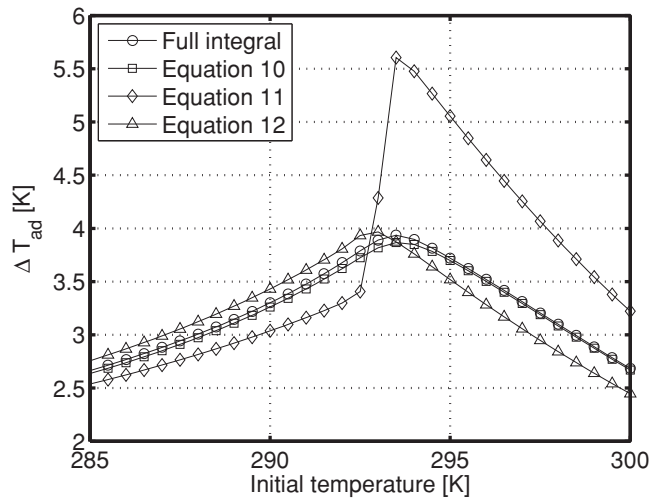


Figure 5. Three approximate expressions for the adiabatic temperature change, compared to an exact calculation ('Full integral'). Data from a mean field model of gadolinium with a field change 0→1 T; experimental results show a similar behavior.

from the functional relationship in Equation 7. This justifies the statement made in the introduction that knowledge of $C_H(T, H)$ permits both ΔS and ΔT_{ad} to be calculated.

4.1. The Maxwell relation

From the expression for the differential of the free energy, F , considered as a function of the temperature, T , and the external field, H , an important relation between the derivatives of S and M may be derived. Considering the external field H as the independent variable we have

$$dF = -SdT - \mu_0 M dH, \quad (14)$$

where μ_0 is the vacuum permeability. Using the fact that the second derivative $\partial^2 F / \partial T \partial H$ is independent of the order of differentiation, we get the so-called Maxwell relation:

$$\left(\frac{\partial S}{\partial H} \right)_T = \mu_0 \left(\frac{\partial M}{\partial T} \right)_H. \quad (15)$$

This equation makes it possible to calculate the isothermal entropy change from the experimentally measured magnetization as a function of temperature and field. Since magnetization measurements in general are much easier to perform than direct measurements of the entropy change, this approach is taken in most of the experimental investigations on magnetocaloric materials. The procedure for the indirect measurement of the magnetocaloric effect has been nicely summarized by Pecharsky and Gschneidner (and is discussed further below).^[50]

However, there has been some controversy in the literature concerning the applicability of the Maxwell relation to first-order magnetocaloric materials.^[53–59] It will therefore be worthwhile to summarize briefly the necessary conditions for the validity of the Maxwell equation. Firstly, and most importantly,

the material is assumed to be in equilibrium. Indeed, this is true of the entire thermodynamic approach taken above. Thus, if there is significant hysteresis, corresponding to a region of metastability of a mixed phase, the Maxwell relation cannot be assumed to be valid, and the results calculated from it may depend on the details of the experimental procedure. This is the origin of the spurious peak sometimes seen in the isothermal entropy change calculated by indirect means.^[59] Secondly, as pointed out by, e.g., A. Giguère et al.,^[53] the Maxwell relation presupposes that the magnetization is a differentiable function of temperature. However, in first-order materials, the magnetization is discontinuous at the transition point, and thus non-differentiable. Instead, Giguère et al. proposed a magnetic analogue of the Clausius-Clayperon equation, involving the finite differences of the magnetization and entropy in the two phases:

$$\left| \frac{T_c(H_f) - T_c(H_i)}{H_f - H_i} \right| = \mu_0 \left| \frac{\Delta M}{\Delta S} \right|, \quad (16)$$

where the transition temperature, T_c , is assumed to shift with the external field. Against this, it may be argued that real-life first-order materials are never strictly discontinuous due to the spreading of the transition caused by, e.g., impurities or local non-stoichiometry.^[58] Only in extremely pure elements can the theoretical zero-width divergence of the heat capacity be observed, e.g. in 99.95 at.% pure Dy.^[60] Furthermore, even with a discontinuous change in entropy, the Maxwell relation can be formally applied, as shown by J.R. Sun et al.^[54]:

$$\begin{aligned} \Delta S &= \mu_0 \int_{H_i}^{H_f} \left(\frac{\partial M}{\partial T} \right)_H dH \\ &= \mu_0 \int_{T_c(H_i)}^{T_c(H_f)} \Delta M \delta(T - T_c) \frac{1}{dT_c/dH} dT_c \\ &\simeq \Delta M \frac{\mu_0(H_f - H_i)}{T_c(H_f) - T_c(H_i)}, \end{aligned} \quad (17)$$

where it has been used that the derivative of M becomes a δ -function and (for the final equality) that dT_c/dH to first order is equal to $(T_c(H_f) - T_c(H_i))/\Delta H$.

Thus, the breakdown of the Maxwell relation is not caused by the first-order nature of a transition in itself. Rather, it is caused by a failure to consider that the sample under certain conditions is in a non-equilibrium state of two coexisting phases and the concomitant history dependence of the state.^[57,61]

5. First-order materials and hysteresis

It is an experimental fact that first-order materials can exhibit a significant hysteresis in both temperature and field. As an example $\text{MnFeP}_{0.59}\text{Si}_{0.30}\text{Ge}_{0.11}$ has a temperature hysteresis of 18 K, as determined from magnetization measurements in heating and cooling experiments.^[62] This means that the (H, T) phase diagram contains two (more or less parallel) lines, one corresponding to decreasing temperature and one corresponding to increasing temperature. The lines intersect the temperature axis in $T_{c\downarrow}$ and $T_{c\uparrow}$, corresponding to the zero-field transition

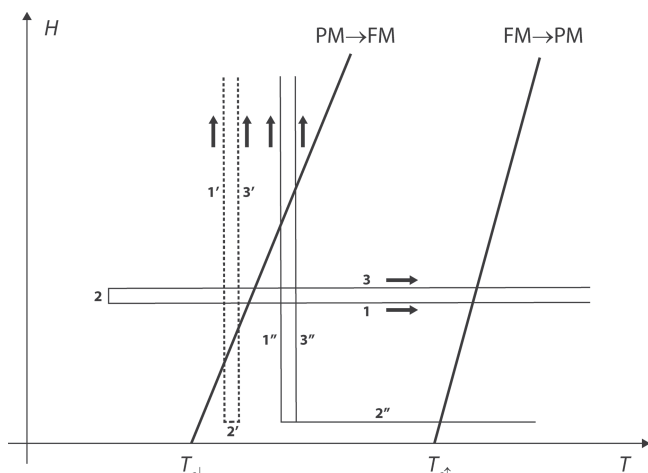


Figure 6. Schematic phase diagram of a first-order material with hysteresis. The high temperature phase is assumed to be paramagnetic (PM), while the low temperature phase is ferromagnetic (FM). Shown are three different paths corresponding to different measurement conditions. The arrows indicate the direction in which measurements are recorded. A) Isofield measurement (1-2-3); B) conventional isothermal measurement (1'-2'-3'); C) 'loop' isothermal measurement (1''-2''-3''), as suggested by Caron et al.^[58] They are discussed further in the text.

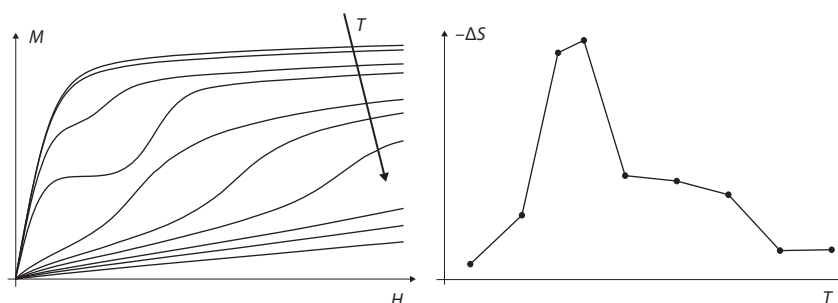


Figure 7. Left: Typical magnetization data obtained by conventional isothermal measurements when measuring a hysteretic first-order material; 10 magnetization curves corresponding to different temperatures are shown. Right: The resulting ΔS curve, as calculated from the Maxwell relation. The spurious spike is caused by the application of equilibrium thermodynamics to a non-equilibrium mixed state.

temperature for decreasing and increasing temperature, respectively. In between these lines the high-temperature and low-temperature phases may co-exist in a metastable state. In Figure 6a schematic phase diagram is shown for a first-order material with a hysteretic transition between a low temperature ferromagnetic (FM) state and a high temperature paramagnetic (PM) state. Experimentally the magnetization may be mapped out in different ways. Let us first consider isofield measurements. These may be performed by starting out with the sample in field at a sufficiently low temperature (below the PM → FM line). Then the temperature is ramped while the magnetization is recorded (cf. Section 9, below). When the high-temperature state is reached, the sample is cooled again, the field incremented, and another temperature ramp performed. From the figure it is clear that at every single instant the sample will either be fully ferromagnetic or fully paramagnetic, justifying the use of the equilibrium thermodynamics above, and in particular the Maxwell relation.

If, on the other hand, isothermal measurements are performed (and being experimentally convenient, they most often are) care has to be taken. The usual procedure is to start in zero field at a given temperature, ramp the field while recording the magnetization, decreasing the field to zero and incrementing the temperature. Outside the region defined by the two lines, the sample will be in a well-defined state throughout the measurement. However, if we start out at a temperature between $T_{c\downarrow}$ and $T_{c\uparrow}$, increasing and then decreasing the field will make the sample first cross the PM → FM line and then re-cross it in the opposite direction; the next measurement is then taken without having crossed the other line (FM → PM), i.e. without completing the ferromagnetic to paramagnetic transition. This means that the sample will be in a mixed, non-equilibrium state when the subsequent measurements are performed. This will result in a set of isothermal magnetization curves exhibiting a series of plateaus or 'knees'. Using the Maxwell relation to calculate the isothermal entropy change from such curves may then result in a spurious spike (see Figure 7). Caron et al. show that this may be avoided if one is careful to cross both lines in the phase diagram, by performing a 'loop' measurement: increasing the temperature between measurements to assure that the fully paramagnetic state is reached.^[58]

It is obviously preferable that measurements are carried out in such a way that non-equilibrium mixed states are not sampled. Still, it is of interest to see to what extent the effect of non-equilibrium states can be corrected for. A number of different authors have considered how to deduce equilibrium values for ΔS from conventional isothermal measurements. Liu et al. subtract part of the area between two magnetization curves to account for the contribution from the mixed state;^[63] this, however, seems to be difficult to justify rigorously.^[58] Das et al. describe the mixed state as consisting of a fraction x of the ferromagnetic state and a fraction $1 - x$ of the paramagnetic state; x is then estimated from the plateaus of the experimental magnetization curves.^[64] Writing the

total magnetization of the mixed state as

$$M = xM_{\text{FM}} + (1 - x)M_{\text{PM}}, \quad (18)$$

where $M_{\text{FM(PM)}}$ is the magnetization of the pure ferromagnetic (paramagnetic) state, they calculate

$$\begin{aligned} \Delta S_{\text{Maxwell}}(T; H_f, H_i) &= \mu_0 \int_{H_i}^{H_f} \left(\frac{\partial M}{\partial T} \right) dH \\ &= \frac{\partial x}{\partial T} \int_{H_i}^{H_f} \mu_0 (M_{\text{FM}} - M_{\text{PM}}) dH \\ &\quad + x\mu_0 \int_{H_i}^{H_f} \left(\frac{\partial M_{\text{FM}}}{\partial T} \right) dH \\ &\quad + (1 - x)\mu_0 \int_{H_i}^{H_f} \left(\frac{\partial M_{\text{PM}}}{\partial T} \right) dH \end{aligned} \quad (19)$$

where it is assumed that H_f is smaller than the critical field at the given temperature (if H_f is larger, there will be an additional term corresponding to the latent heat of the transition). The two last terms in Equation 19 are simply the total equilibrium contribution from the ferromagnetic and paramagnetic phase fractions. Das et al. identify the first term, $(\partial x/dT) \int_0^{H_f} (M_{FM} - M_{PM}) dH$, as the main non-equilibrium contribution which should be discarded. They show that this procedure leads to reasonable results, both experimentally and in a simple model.

Another possible reason for deviations from the Maxwell relation is domain effects. In zero field the total magnetization of the sample will be zero, even in the ferromagnetic state where the intrinsic magnetization is non-zero. This is due to the domain structure which is created within the sample to minimize the magnetic self-energy. As a field is applied, the domains will align to the field, giving a rapid increase in total magnetization which does not reflect the change in intrinsic magnetization. This will lead to an overestimation of the magnetocaloric effect.^[57,65] However, most magnetocaloric materials are magnetically soft and will have the domains aligned fully at relatively low fields, especially just below the Curie temperature where the intrinsic magnetization causing the domain structure is small; once the domains are aligned, the above analysis becomes valid again.

Above, we have considered the behavior of the phase diagrams as being governed by two degrees of freedom, H and T . Recently, Gama et al. argued that the first-order transition in magnetic materials such as MnAs or $Gd_5Ge_2Si_2$ should be considered as governed not by two thermodynamic degrees of freedom (temperature and field) but by three; then, according to the Gibbs phase rule, instead of a single point there will be a finite region of the phase diagram where two phases can coexist.^[66] Thus, the width of the transition should not be understood as merely due to non-ideal effects such as disorder but as due to a third degree of freedom in the thermodynamic system. Although this degree of freedom is not explicitly identified by Gama et al., they show that some experimental features of the transition have a natural explanation within this framework. If their interpretation is correct, it would seem to imply that there is an intrinsic limit to how much the hysteresis of such a material can be reduced.

6. Theoretical approaches

A theoretical understanding of the magnetocaloric effect must combine two subjects challenging in themselves: magnetism and structural phenomena. The crucial point in any microscopic model is how to include the spin-lattice interaction responsible for the magnetocaloric effect. The basic theoretical framework is well-known, being that of magnon-phonon interactions, as summarized by, e.g., Jensen and Mackintosh.^[67] However, calculations with the full magnon-phonon interaction (or indeed realistic phonon dispersions) is seldom done in any realistic detail due to the computational complexities; complexities that for first-order materials are compounded by the necessity of including the non-linear terms of the interaction in order to have an isothermal entropy change which is not solely given

by the magnetic part of the system. Instead the usual approach is to study simple model systems. Even these will usually have to be solved approximately. Here we review some of the more widely used approaches in the literature.

6.1. Landau theory

Landau theory is a useful starting point for the discussion of both second-order and first-order materials. It assumes the existence of a macroscopic order parameter (in the present case the intrinsic magnetization) which is non-zero in the low-temperature, ordered phase and zero in the high-temperature, disordered phase. The ordinary Landau theory ignores fluctuations in the order parameter which certainly play a role in real systems, and is in this respect similar to the mean field theories discussed below. In particular, Landau theory predicts the wrong critical exponents around T_c . Nevertheless, it is widely used and can in many cases give a qualitative understanding.

The basic assumption of Landau theory is that close to the critical temperature, the free energy may be expanded in powers of the intrinsic magnetization:

$$F(T, M) = F_0(T) + a(T - T_0)M^2 + bM^4 + cM^6 - \mu_0 MH, \quad (20)$$

where terms to the sixth order have been kept and the Landau parameters a and c are greater than zero. If also $b > 0$, the transition is of second order and the M^6 term may be dropped. In this case the equilibrium condition $\partial F/\partial M = 0$ becomes

$$2a(T - T_0)M + 4bM^3 - \mu_0 H = 0. \quad (21)$$

This cubic equation may be solved to give M as a function of T and H . A non-zero solution for $H = 0$ only exists for $T < T_0$, i.e. in this case we have $T_c = T_0$. The entropy, $S = -(\partial F/\partial T)_H$, becomes $S = S_0(T) - aM^2$, where S_0 is the non-magnetic part of the entropy related to phonons and electrons (phonon-magnon interactions will give rise to temperature dependence of the Landau parameters which for the moment is ignored). The isothermal entropy change is

$$\Delta S(T; H_f, H_i) = -a(M(T, H_f)^2 - M(T, H_i)^2). \quad (22)$$

Bashkatov and Bodryakov use these expressions together with the approximate expression Equation 10 for the adiabatic temperature change to investigate the magnetocaloric effect within Landau theory.^[68] From Equation 22 it can be shown that $-\Delta S$ always peaks at $T = T_c$. However, this is not true of ΔT_{ad} even within Landau theory. The authors find that the peak temperature of ΔT_{ad} shifts slightly up with increasing field; for a gadolinium-like material they find, e.g., a shift of 2.4 K in a field of 4 T. They further investigate the optimum magnetocaloric performance in terms of the Landau parameters a and b .

Turning now to the case $b < 0$, the transition temperature becomes $T_c = T_0 + \frac{b^2}{4ac}$ at which point there will be a finite jump in magnetization of magnitude $(\frac{-b}{2c})^{1/2}$, i.e. the transition will be of first order. Several authors have studied first-order materials within this framework. Amaral and Amaral find that

in this case the peak entropy change shifts to higher temperatures with field, in accordance with experiment, and that the form of the ΔS curve depends on the temperature dependence of the Landau parameter b .^[69] Carvalho et al. study the magnetic hysteresis in MnAs.^[70] For the first-order transition Landau theory predicts two minima of M , one of which is metastable. This gives rise to hysteresis, and the authors find a qualitative agreement with experiment, including a spurious peak in ΔS when calculated directly from the Maxwell relation. Yamada and Goto use a phenomenological Ginzburg-Landau theory to include spin fluctuations in the Landau theory; these give rise to a temperature dependent renormalization of the Landau parameters.^[71] Yamada and Goto show that the isothermal entropy change in this case depends not only on the jump in magnetization at T_c but also on the temperature derivative of the critical field. The same framework is used by Fujita and Fukamichi to study the magnetocaloric effect of $\text{La}(\text{Fe}_x\text{Si}_{1-x})_{13}$.^[72]

6.2. Molecular mean field theory

The molecular mean field theory simply approximates the system of interacting spins with a collection of non-interacting spins in a field determined by the average spin value.^[73] The magnetization then becomes

$$M = N_s g J \mu_B B_J(\chi), \quad (23)$$

where N_s is the number of spins per unit mass, g is the Landé factor, J is the quantum number of a single spin, and μ_B is the Bohr magneton. The parameter χ has to be found self-consistently from the set of coupled equations involving the Brillouin function, B_J :

$$B_J(\chi) = \frac{2J+1}{2J} \coth\left(\frac{2J+1}{2J}\chi\right) - \frac{1}{2J} \coth\left(\frac{1}{2J}\chi\right) \quad (24)$$

$$\chi = \frac{gJ\mu_B\mu_0 H}{k_B T} + \frac{3T_c J}{T(J+1)} B_J(\chi). \quad (25)$$

Here k_B is Boltzmann's constant. The magnetic entropy is readily found from the partition function for independent spins. Using the Debye expression for the lattice entropy and the Sommerfeld expression for the conduction electrons allows one to make a complete evaluation of the isothermal entropy change and the adiabatic temperature change. Oesterreicher and Parker investigated the magnetocaloric effect within this framework.^[74] In the molecular mean field model the peak isothermal entropy change occurs at $T = T_c$ while the peak adiabatic temperature change is shifted to a slightly higher temperature, similarly to the Landau theory. This is unsurprising, given that they are both mean field theories.

Although the shortcomings of the molecular mean field model are well-known,^[75] it has the great advantage of allowing a consistent set of thermodynamic curves to be calculated. For this reason, it is widely used for modeling purposes.^[76] It can also offer a useful framework for the interpretation of experimental results.^[77] And for some second-order materials, notably gadolinium, the correspondence of the mean field theory to experiment is actually quite satisfactory, although it is unable to

account for the weak displacement of the peak temperature of ΔS with increasing field.

6.3. Bean-Rodbell like models

A straightforward extension of the molecular mean field theory is to consider a collection of spins which interact through an exchange coupling the strength of which depends linearly on the atomic spacing between them. Such a model was first considered by Bean and Rodbell in an investigation of the first-order transition in MnAs.^[78] Since the exchange coupling and the Curie temperature are proportional in the mean field model, their assumption can be expressed as

$$T_c = T_0 \left(1 + \beta \frac{v - v_0}{v_0}\right), \quad (26)$$

where T_0 is the Curie temperature if the lattice were incompressible, v is the volume, and v_0 is the volume in the absence of exchange interactions; the parameter β may be positive or negative. Bean and Rodbell showed that the phase transition changes from ordinary second order to first order, according to whether

$$\eta = A_J N K k_B T_0 \beta^2 \quad (27)$$

is smaller or greater than 1. Here K is the compressibility and N is the number of spins per unit volume, while A_J is a numerical factor which depends on the spin of the particles; for the case considered by Bean and Rodbell ($J = \frac{1}{2}$) it equals $\frac{3}{2}$.

Tegus et al. applied the Bean-Rodbell model to the first-order phase transition in $\text{MnFeP}_{1-x}\text{As}_x$, using $J = 2$.^[79] They show that by combining the Bean-Rodbell model with a Landau free energy approach one can gain a qualitative understanding of the field-induced metamagnetic transition. Jia et al. used a similar approach to study the magnetovolume effects caused by the introduction of interstitial hydrogen in $\text{La}(\text{Fe}_x\text{Si}_{1-x})_{13}$.^[80] Valiev analyzed the general dependence of the magnetocaloric effect on the magnetoelastic coupling, showing that it increases with increasing coupling.^[81] The analysis was then applied to two different compositions of $\text{La}(\text{Fe}_x\text{Si}_{1-x})_{13}$ ($x = 0.86, 0.88$).^[82]

6.4. Model Hamiltonians

To describe the magnetocaloric effect of specific materials in more detail the usual approach is to consider model Hamiltonians which capture the physical effects expected to be most important. The main question is how to treat the magnetism which can either be due to localized spins or to itinerant electrons. Thus, in rare earth compounds where the magnetism arises from localized spins, the Heisenberg Hamiltonian is a natural starting point:

$$H_{\text{mag}} = -\frac{1}{2} \sum_{i,j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j - g\mu_0 \mu_B \sum_i s_{i,z} H, \quad (28)$$

where i, j range over all spin sites and $\mathbf{s}_i$ is the spin at site i ; the external field H is assumed to be along the z -direction. A crystal field term may be added to Equation 28, removing the

zero field degeneracy of the spin states of a single site. For the phonons, the following Hamiltonian is used

$$H_{\text{ph}} = \sum_{\mathbf{q}} \hbar \omega_{\mathbf{q}} a_{\mathbf{q}}^{\dagger} a_{\mathbf{q}}, \quad (29)$$

where $a_{\mathbf{q}}^{\dagger}$ and $a_{\mathbf{q}}$ are phonon creation and annihilation operators, and $\omega_{\mathbf{q}}$ is the phonon dispersion relation. The exchange interactions J_{ij} should in principle depend on the lattice spacing to account for the magnetoelastic coupling; this will then introduce terms involving both spin and phonon operators in the total Hamiltonian $H = H_{\text{mag}} + H_{\text{ph}}$,^[83] complicating things greatly. Instead it is usually assumed that the main effect of the coupling will be to renormalize the phonon frequencies and to introduce a quartic term in the spin operators. The resulting Hamiltonian can either be solved in a mean field approximation,^[56] or using numerical methods such as Monte Carlo simulations.^[84,85] De Oliveira extended the model to include disorder effects,^[86] while Santana et al. studied the effects of hysteresis within the mean field approximation.^[87]

A similar approach, but involving two sublattices with different exchange coupling, has been used to investigate the inverse magnetocaloric effect in antiferromagnets and ferromagnets.^[39] Schmidt et al. studied the frustrated $J_1 - J_2$ model in 2D using both mean field approximations and exact diagonalization of finite clusters combined with the finite temperature Lanczos method.^[88]

For transition metal based compounds where itinerant electrons cause the magnetism, Hubbard-like models have been proposed.^[89,90]

$$H_{\text{mag}} = \sum_{i,j,\sigma} t_{ij} c_{i\sigma}^{\dagger} c_{j\sigma} + U \sum_i (c_{i\uparrow}^{\dagger} c_{i\uparrow})(c_{i\downarrow}^{\dagger} c_{i\downarrow}) + g \mu_0 \mu_B H \sum_{i,\sigma} \sigma (c_{i\sigma}^{\dagger} c_{i\sigma}). \quad (30)$$

Here the first term describes the electron hopping with an amplitude t_{ij} ($c_{i\sigma}^{\dagger}$ and $c_{i\sigma}$ being electron creation and annihilation operators), while the second term describes on-site repulsion between electrons with a strength U and the last term is the coupling to the external field H . To this Hamiltonian is then added the phonon Hamiltonian (Equation 29) and a term describing the electron-phonon interaction. Again, it is usually assumed that the main effect of the latter will be to renormalize the phonon frequencies. Within the mean field approximation it is then possible to give a complete calculation of the magnetocaloric effect.^[89]

The application of the above models to different magnetocaloric systems has recently been extensively reviewed by de Oliveira and von Ranke.^[91]

6.5. First-principles calculations

There have been a few attempts on first-principles studies of the magnon-phonon interaction in general: Sabiryanov and Jaswal studied magnon-phonon interactions in iron and found that phonons only have minor effects on the magnons,^[92] while Fennie and Rabbe studied the influence of magnons on phonon frequencies in ZnCr_2O_4 .^[93]

Specific studies of the magnetocaloric effect using density-functional theory to calculate the electronic structure have also been published. Kuz'min and Richter investigated $\text{La}(\text{Fe}_x\text{Si}_{1-x})_{13}$ and showed that the energy vs. magnetization curve is very flat with shallow minima and that this is the reason for the low hysteresis in this system.^[94] Buchelnikov et al. used first-principles calculations of the exchange couplings in $\text{Ni}_2 + x\text{Mn}_{1-x}\text{Ga}$ in combination with Monte Carlo simulations to investigate the first-order transition in this Heusler alloy.^[95] Gercsi and Sandeman used density-functional theory on MnP type compounds and showed that Mn-Mn distances play a crucial role for determining the type of magnetic order.^[96] This study was extended by Gercsi et al. and used as the basis for designing a new series of alloys, $\text{CoMnGe}_{1-x}\text{P}_x$, which were then synthesized and investigated experimentally.^[97]

These studies clearly show that first-principles calculations can play a very fruitful role in the theoretical understanding of the magnetocaloric effect.

6.6. Hysteresis modeling

As described above, hysteresis can be considered within the framework of Landau theory and mean field theories. However, since hysteresis is a manifestation of a system far from equilibrium it should in principle be described using non-equilibrium thermodynamics.^[98] In several papers Basso and coworkers have pioneered the use of non-equilibrium thermodynamics on magnetocaloric systems.^[99–103] The irreversible entropy generation is described by introducing a set of additional, internal state variables which do not couple to any intensive variables. The internal variables in principle describe the non-equilibrium magnetization state. The point of introducing the internal state variables is that one can write down an out-of-equilibrium free energy as a function of both the usual (external) state variables and the internal variables. To make progress, the description is drastically simplified by modeling the system as a collection of independent bistable units. Each unit can have magnetization $\pm \Delta m$ with corresponding energies $E_{\pm} = \pm \mu_0 \Delta m (h_u - H)$ separated by an energy barrier $\mu_0 \Delta m h_c$. This means that the external field H will be able to change the $-\Delta m$ state to $+\Delta m$ when increasing above $h_u + h_c$ while the $+\Delta m$ state will flip to $-\Delta m$ when the external field decreases below $h_u - h_c$. This is evidently an elementary hysteresis loop. The total system is characterized by the probability density function $p(h_c, h_u)$ known as the Preisach distribution, while a given (possibly non-equilibrium) state is characterized by a line in the $h_u - h_c$ -plane separating the units which are in the $+\Delta m$ state from the ones in the $-\Delta m$ state. Both h_u , h_c and Δm will be temperature dependent. However, Basso and coworkers only consider the case where they may be rescaled to temperature-independent variables. The free energy becomes an integral of the energies weighted by the Preisach distribution, and from this it is possible to calculate the magnetization and entropy.

This approach allows an interesting insight into the generic mechanisms of hysteresis and the approach to equilibrium, although at present it seems difficult to make contact with the concrete microscopics of a given material.

7. Scaling of the magnetocaloric effect

When the isothermal entropy change or adiabatic temperature change of a given material in different external fields is mapped out as a function of temperature, a series of curves with roughly similar appearance is generated. Each curve has a peak near the Curie temperature and falls off more or less smoothly on each side. Both the peak height (i.e. $(-\Delta S)_{\max}$ or $\Delta T_{\text{ad, max}}$ for the ordinary magnetocaloric effect), the peak temperature and the width of the peak in general depend on the external field strength. The scaling behavior of these curves has been studied experimentally and theoretically by several authors.

As previously mentioned the maximum isothermal entropy change, $(-\Delta S)_{\max}$ experimentally shows a power law dependence on the external field (at least for a range of fields):

$$(-\Delta S)_{\max} \propto H^n, \quad (31)$$

where n is smaller than 1. Oesterreicher and Parker found that within the mean field model as applied to second-order materials this exponent equals $2/3$,^[74] which indeed is close to that observed experimentally for, e.g., Gd.^[10] Within the mean field model it is also found that the peak temperature is equal to the Curie temperature, independent of the field. Going beyond mean field theories, Franco et al. showed, using scaling arguments valid for critical phenomena close to the critical temperature, that the exponent at the peak temperature in general changes,^[104] while the peak temperature becomes a function of field and moves away from the Curie temperature according to a power law.^[105] However, the changes are relatively small, typically less than 10% for the case of n ,^[104] necessitating careful experiments to distinguish them.

Two further temperature ranges exist where a scaling exponent can be identified: For $T \ll T_c$ the intrinsic magnetization becomes practically field independent, giving rise to an exponent of $n = 1$ by the use of the Maxwell relation, Equation 15; for $T \gg T_c$ the magnetization can be assumed to follow the Curie-Weiss law, $M \propto H/(T - T_0)$, giving rise to $n = 2$.^[104,106] Inspired by this, one can investigate a general scaling relation of the form

$$-\Delta S(T; H, 0) = \alpha(T) H^{n(T)}, \quad (32)$$

where the prefactor $\alpha(T)$ and the exponent $n(T)$ depend on temperature but not on field. Experimentally this seems to be a reasonable interpolation form for the isothermal entropy change, although some sensitivity to the field is seen when fitting $\alpha(T)$ and $n(T)$ from data.

Based on scaling ideas, Franco and coworkers have proposed a phenomenological rescaling procedure for both the ΔS and ΔT_{ad} curves.^[106] Given a family of ΔS curves for a number of different fields, they are all normalized by the peak value, and a rescaled temperature θ is defined as

$$\theta = \begin{cases} \frac{T - T_{\text{peak}}(H)}{T_{\text{peak}}(H) - T_{r1}(H)} & \text{if } T \leq T_{\text{peak}}(H) \\ \frac{T - T_{\text{peak}}(H)}{T_{r2}(H) - T_{\text{peak}}(H)} & \text{if } T > T_{\text{peak}}(H) \end{cases} \quad (33)$$

where $T_{r1}(H)$ ($T_{r2}(H)$) is the temperature below (above) $T_{\text{peak}}(H)$ at which ΔS has fallen to a fraction r of its peak value.^[104,107] (The use of a single T_r , both above and below the peak

temperature has also been considered.^[106] The resulting curves $\Delta \tilde{S}(\theta, H)$ all pass through the same three points $(-1, r)$, $(0, 1)$ and $(1, r)$. Franco and coworkers have applied this framework to a large number of materials, including Gd,^[108] $(\text{Er}_{1-x}\text{Dy}_x)\text{Al}_2$,^[109] FeMoCuB amorphous alloys,^[110] Hitperm- and Nanoperm-type alloys,^[111,112] and Fe-B-Cr-based amorphous alloys,^[113] showing that in many cases the rescaled curves for a given material collapse onto a single curve. This is also true if the effects of demagnetization^[114] and minority phases^[115] are considered. One can also allow the exponent n in Equation 32 to depend not only on T but also on H by defining it as $n(T, H) = -d \ln \Delta S / d \ln H$; n may then be rescaled using the same approach as for ΔS .^[106] A theoretical justification for the collapse has been attempted using scaling theory.^[116,117] However, it may be questioned whether scaling relations, valid only in the critical region close to T_c (where the curves by construction must be very close together), can be used to explain the collapse of curves over very wide temperature ranges.

Turning now to the scaling properties of the adiabatic temperature change, it is seen from Equation 11 that if the field dependence of $C(T, H)$ is ignored ΔT_{ad} will scale with field in the same way as ΔS . Of course, the specific heat does depend on field, and this has implications for the scaling of ΔT_{ad} , even in the mean field theory. One consequence, already noted by Oesterreicher and Parker,^[74] is that the peak temperature of ΔT_{ad} shifts slightly towards higher temperatures with increasing field. Another consequence is that the scaling exponent at the peak temperature is not $2/3$ but slightly higher; a numerical study found an exponent of 0.68 for Gd.^[118] Some authors have come to a different conclusion, claiming that the scaling of $(-\Delta S)_{\max}$ and $\Delta T_{\text{ad, max}}$ are exactly identical in the mean field model.^[108] This, however, is due to the fact that they only consider the singular part of the specific heat in their derivation. In reality, a sizable non-universal (H -independent) background due to phonons is superimposed on the peak in the specific heat, influencing the scaling behavior.

The scaling relation for the maximum isothermal entropy change, Equation 31, is experimentally well verified for intermediate fields. However, it turns out that a better fit to the data is achieved if one allows for a slight constant offset: $(-\Delta S)_{\max} = A H^n - \text{const.}$ ^[119] Kuz'min et al. have showed that within Landau theory this fact can be explained by the presence of impurities which have the effect of introducing a spatial variation of the Curie temperature of the sample.^[120] Since ΔS obviously has to go to zero for $H \rightarrow 0$ this expression has to be modified at small H , while at large H the effects of saturation will limit the increase of $-\Delta S$ (saturation will start to play a role when $\mu_B \mu_0 H \sim k_B T$). Taking this into account Lyubina et al. extended the analysis to arrive at the expression

$$(-\Delta S)_{\max} = A(H + H_0)^{2/3} - A H_0^{2/3} + B H^{4/3}, \quad (34)$$

valid within Landau theory,^[121] A and B are positive constants. A similar expression for ΔT_{max} is found:^[122]

$$\Delta T_{\text{max}} = A'(H + H_0)^{2/3} - A' H_0^{2/3} + B' H^{4/3} \quad (35)$$

with different constants A' , B' . In both cases the $H^{4/3}$ term can be ignored if saturation conditions are not approached. Bahl et al. have shown that within the molecular mean field

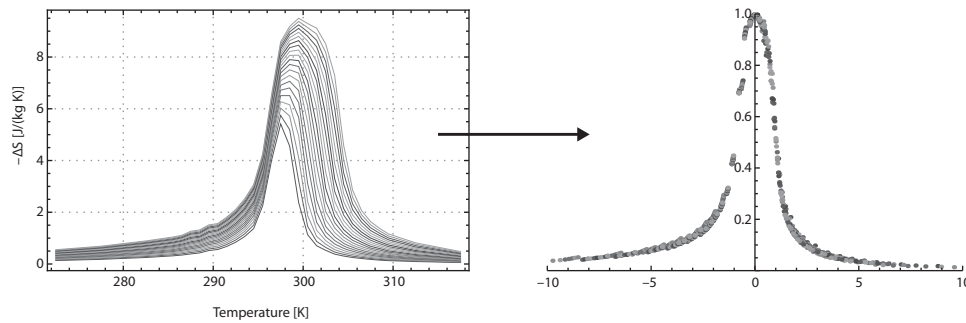


Figure 8. Left: The negative of the isothermal entropy change in a $\text{MnFeP}_{1-x}\text{As}_x$ sample with a transition temperature of 298 K and an applied field ranging from 0.1 to 1.5 T. The curves were derived from isothermal magnetization measurements. Right: Entropy curves rescaled using Equation 33. Although the material is first order, a collapse to a single curve is seen.^[125]

model a distribution of Curie temperatures has the additional effects of changing the peak temperature and field dependence of both the specific heat, the isothermal entropy change, and the adiabatic temperature change.^[123] Experimentally it will be challenging to unequivocally disentangle the effects of non-mean-field behavior and impurities.

The scaling of first-order materials has not been studied as extensively as that of second-order materials. Bonilla et al. propose as an experimental criterion of a first-order transition that the corresponding entropy curves do not collapse upon rescaling.^[124] However, the entropy curves of some first-order materials such as $\text{MnFeP}_{1-x}\text{As}_x$ can be collapsed (see **Figure 8**) so further investigations are needed on this point.^[125]

8. Fundamental properties of ΔS and ΔT_{ad}

Independent of the above phenomenological scaling considerations it is of interest to investigate what properties of ΔS and ΔT can be deduced from fundamental principles.

Since $\Delta S(T; H_f, H_i)$ gives the isothermal entropy change when the magnetic field is changed from H_i to H_f , it follows directly that for reversible materials the following relation must hold:

$$\Delta S(T; H_f, H_i) = -\Delta S(T; H_i, H_f). \quad (36)$$

Thus, the magnitude of the demagnetization and magnetization entropy change peaks at the same temperature. As we have seen above, this is a temperature close to, but not necessarily equal to, T_c .

Often the total entropy is written as a sum of ‘magnetic’, ‘lattice’, and ‘electronic’ entropies, where it is assumed that the latter two do not depend on the field:^[49]

$$S(T, H) = S_{\text{mag}}(T, H) + S_{\text{lat}}(T) + S_{\text{el}}(T). \quad (37)$$

S_{mag} is then identified with the entropy of the spin system, S_{lat} with the entropy associated with the lattice vibrations (phonons), and S_{el} with that of the conduction electrons. From this it is inferred that the isothermal entropy change is equal to the change in ‘magnetic’ entropy: $\Delta S(T; H_f, H_i) = S_{\text{mag}}(T, H_f) - S_{\text{mag}}(T, H_i)$. For many second-order materials, in particular for Gd, this is indeed true and allows one to derive a fundamental limit on the maximum entropy change. This is done by

considering the paramagnetic state as a collection of independent, localized spins with magnitude J (in units of $\hbar$), disregarding interactions and any anisotropy. In zero field, each of the $(2J + 1)$ possible orientations of the spin are equally probable, giving an entropy per spin of $k_B \ln(2J + 1)$. In very high fields ($\mu_B \mu_0 H \gg k_B T$), all spins are aligned to the field direction, giving an entropy of 0. Thus, the maximum achievable change in magnetic entropy is

$$(-\Delta S)_{\text{max}}(T; H \rightarrow \infty, 0) = Nk_B \ln(2J + 1), \quad (38)$$

where N is the number of spins per unit mass in the system. For finite field changes this number is reduced, as it is in the ferromagnetic state where there is a net magnetization. Therefore the above equation is a strict limit which will not be attained in actual ferromagnetic materials. For Gd (having $J = 7/2$ and an atomic weight of 157) we get $(-\Delta S)_{\text{max}} = 110 \text{ J/kg} \cdot \text{K}$, i.e. an order of magnitude higher than typical experimental values. This shows that that for realistically attainable fields the theoretical value from Equation 38 is of no relevance.

The above splitting of the total entropy into different terms cannot be thermodynamically justified when there is a strong coupling between the different subsystems. In that case the above analysis becomes inapplicable. For first-order materials, where the magnetocaloric effect often is caused by a strongly non-linear coupling of the spin and lattice degrees, an isothermal entropy change exceeding the above $(-\Delta S)_{\text{max}}$ has indeed been observed experimentally.^[126–128] Furthermore, on the theoretical side Mukherjee et al. demonstrate this point convincingly by setting up a minimal model for interacting classical Ising spins incorporating a non-linear spin-lattice interaction and showing that the corresponding entropy change indeed can exceed that given by Equation 38.^[129] Basso comes to a similar conclusion, using a Landau free energy approach similar to that described in Section 6.3 to demonstrate that the total entropy change may either be the sum or the difference of magnetic and structural contributions; in the first case the entropy change obviously exceeds the magnetic limit.^[130] This also implies that it is not *a priori* obvious if there exists a fundamental limit for the isothermal entropy change in actual materials.

Of course one could allow all three terms in Equation 37 to depend on field:

$$S(T, H) = S_{\text{mag}}(T, H) + S_{\text{lat}}(T, H) + S_{\text{el}}(T, H). \quad (39)$$

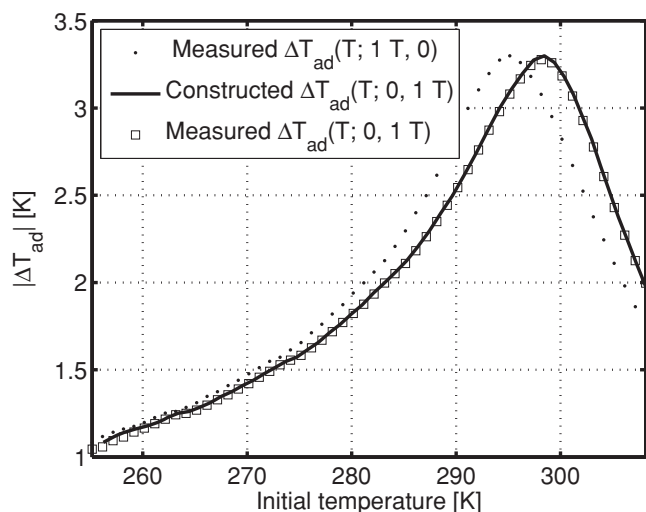


Figure 9. Magnetization ($\Delta T_{\text{ad}}(T; H, 0)$) and demagnetization ($\Delta T_{\text{ad}}(T; 0, H)$) curves for the adiabatic temperature change in commercial grade gadolinium for $H = 1$ T, determined by direct measurement. The two curves are displaced with respect to one another and do not have the same form. This is a direct consequence of the definition of ΔT_{ad} . The full line shows $\Delta T_{\text{ad}}(T; 0, H)$ calculated from the measured $\Delta T_{\text{ad}}(T; H, 0)$ and Equation 40.

This can be convenient for modeling purposes where the calculation of the entropy may involve several terms which can be identified as coming primarily from the spins, the lattice or the itinerant electrons. However, it should be kept in mind that the splitting is in that case to some extent merely a convention. In particular, the Maxwell relation (when applicable) relates the field derivative of the *total* entropy (not just the ‘magnetic’ part) to the temperature derivative of the magnetization.

For ΔT_{ad} the relation between the temperature change upon magnetization and demagnetization, respectively, is

$$\Delta T_{\text{ad}}(T; H_f, H_i) = -\Delta T_{\text{ad}}(T + \Delta T_{\text{ad}}(T; H_f, H_i); H_i, H_f). \quad (40)$$

This is simply a statement of the fact that the system upon adiabatic magnetization from H_i to H_f followed by adiabatic demagnetization from H_f to H_i will have a final temperature equal to the starting temperature. It immediately follows that $|\Delta T_{\text{ad}}(T; H_f, H_i)|$ and $|\Delta T_{\text{ad}}(T; H_i, H_f)|$ do not peak at the same temperature (see Figure 9), a point which is not always appreciated in the literature. It is also seen that the approximate relations derived above, Equations 10–12, do not obey this functional relationship. When modeling the magnetocaloric process in, e.g., an active magnetic regenerator, it is important to use a functional form of ΔT_{ad} which is consistent with Equation 40 to ensure that the energy balance is not violated.^[131] From Equation 40 together with the assumption that ΔT_{ad} is a continuous function of temperature it is, as shown by Nielsen et al.^[132] possible to derive a universal limit on the variation of ΔT_{ad} with temperature:

$$\frac{d\Delta T_{\text{ad}}(T; H_f, H_i)}{dT} \geq -1. \quad (41)$$

For irreversible materials the equality, Equation 36, now becomes an inequality due to the irreversible generation of

entropy which will have taken place during the cycle $H_i \rightarrow H_f \rightarrow H_i$:

$$\Delta S(T; H_f, H_i) > -\Delta S(T; H_i, H_f). \quad (42)$$

For the adiabatic temperature change, the corresponding relation becomes

$$\Delta T_{\text{ad}}(T; H_f, H_i) > -\Delta T_{\text{ad}}(T + \Delta T_{\text{ad}}(T; H_f, H_i); H_i, H_f). \quad (43)$$

Nielsen et al.^[132] show that in this case, the strict inequality, Equation 41, is relaxed to

$$\frac{d\Delta T_{\text{ad}}(T; H_f, H_i)}{dT} \geq -1. \quad (44)$$

Having derived a constraint on the derivative of ΔT_{ad} it is natural to consider whether a limit on the adiabatic temperature change itself may be derived. If one were naively to use the theoretical $(-\Delta S)_{\text{max}}$ from Equation 38 together with the approximate expression for ΔT_{ad} applicable to second-order materials, Equation 12, one would get for gadolinium (using a specific heat capacity of ~ 300 J/kg·K) that the maximum achievable adiabatic temperature change in the vicinity of the Curie point would be on the order of 100 K! In a recent paper, Zverev et al. have attempted to derive a more realistic limit on the maximum temperature change for finite field changes.^[133] Their argument takes its starting point in Equation 4, expressing the infinitesimal temperature change dT upon a field change dH in terms of T , C_H and $(\partial S/\partial H)_T$. From this equation it immediately follows that outside the interval where $(\partial S/\partial H)_T = \mu_0(\partial M/\partial T)_H$ is non-zero (<0 , actually), the adiabatic temperature change cannot increase further. Thus, a material with a high but narrow peak in $(\partial S/\partial H)_T$ has its adiabatic temperature change determined not by the height of the peak but by the width of the interval where the derivative is non-zero. On the other hand, if the peak is low but wide, the adiabatic temperature will be limited by the peak height. Zverev et al. proceed with some highly simplifying assumptions: 1) that $(\partial M/\partial T)$ is constant, independent of T and H , in a temperature interval δ in the vicinity of the Curie temperature; its value is then $-\Delta M/\delta$ where ΔM is the difference in magnetization from the low temperature side to the high temperature side of the interval, and 2) that in this temperature interval T/C_H can also be considered as a constant. The maximum ΔM is equal to the saturation magnetization of the given compound which will be limited by the maximum magnetization of its constituent elements. Of the elements in the periodic table Dy and Ho have the highest atomic magnetization, about $10 \mu_B$ per atom, where μ_B is the Bohr magneton. This limit on ΔM implies that the area below the ΔS curve is also limited, being equal to $\mu_0 \Delta M H$. Under the above assumptions, the adiabatic temperature change will be given as $\Delta T_{\text{ad}} = (T/C_H)(\Delta M/\delta)\mu_0 H$ when the peak is sufficiently wide. From this it is seen that decreasing the peak width (keeping the area constant) will increase the adiabatic temperature change until the point where it becomes limited by the width of the peak and can increase no further. Evidently this happens when

$$\Delta T_{\text{ad}} = \frac{T}{C_H} \frac{\Delta M}{\delta} \mu_0 H = \delta. \quad (45)$$

Thus, under these assumptions, the maximum attainable adiabatic temperature change is

$$\Delta T_{\text{ad,max}} \sim \sqrt{\frac{T}{C_H}} \Delta M(\mu_0 H). \quad (46)$$

For Gd at 300 K ($T/C_H = 1 \text{ K}^2\text{kg/J}$, $\Delta M = 7 \mu_B/\text{atom}$) the limit becomes $\Delta T_{\text{ad,max(Gd)}} \sim 16\sqrt{\mu_0 H/(1 \text{ T})} \text{ K}$. No other ferromagnetic element with a transition near room temperature exists. Instead one may consider a hypothetical binary composition with $T_C = 300 \text{ K}$, containing a nonmagnetic element and an element with a maximal magnetization of $10 \mu_B$. Then, with $\Delta M = 10 \mu_B$ per formula unit and C given by the Dulong-Petit law, the limit becomes $\Delta T_{\text{ad,max(binary)}} \sim 18\sqrt{\mu_0 H/(1 \text{ T})} \text{ K}$. Real materials will obviously perform much worse than this. On the other hand, the argument neglects the possible entropy contributions from non-linear spin-lattice interactions and gives the wrong field dependence of $\Delta T_{\text{ad,max}}$ (an exponent of $\frac{1}{2}$ rather than the $\frac{2}{3}$ of Equation 35). A more realistic model would be needed to address these issues.

9. Characterization methods

Turning now to experiment, in this section we review some widely used methods to characterize magnetocaloric materials experimentally. These include magnetization measurements, direct and indirect measurements of the adiabatic temperature change, measurements of the field dependent specific heat, and direct measurements of the isothermal entropy change. For all of these methods it is important to take magnetostatic demagnetization effects properly into account, and the final part of this section takes up the effects of demagnetization on characterization in a little more detail.

9.1. Magnetization measurements

By far the most common method for measuring ΔS is to map out the applied magnetic field and temperature dependence of the magnetization of the sample, $M(H, T)$, which with the use of the Maxwell relation (Equation 15) can be used to calculate ΔS . Equipment for measuring the temperature and magnetic field dependence of M is readily available in many laboratories. Applied fields ranging up to about 10 T in broad temperature ranges down to a few K are available, but this requires helium cooled superconducting magnets and measurements are often time consuming. Within the ranges relevant for room temperature magnetic refrigeration devices, i.e. a temperature range of about 200–350 K and an applied magnetic field range of 0–2 T, a conventional vibrating sample magnetometer (VSM) using electromagnets is well suited. Here the magnetic moment of a sample is determined by measuring the voltage induced in pickup coils when vibrating the sample in a constant homogeneous magnetic field. The setup is constructed so that the temperature of the sample may be controlled and recorded. The advantage is that the moment of any sample shape (bulk, plate, fragments, powder, etc.) in quantities ranging from below one and up to a few tens of cubic millimeters may be measured. Dividing

with the sample mass gives the specific magnetization, useful for calculating ΔS per unit mass. The entropy change per unit volume is usually found from ΔS per unit mass using the bulk sample density, as the actual sample volume may be hard to measure.

When mapping out $M(T, H)$ two basic routes may, as already mentioned, be followed: either keeping the applied field constant and varying the temperature, or keeping the temperature constant, varying the applied field. In the first method $\partial M/\partial T$ may be found directly and numerically integrated to find ΔS . In the second approach the area between two magnetization isotherms is integrated to give ΔS at the intermediate temperature. It is straightforward to derive an estimate of the magnitude of the error in the calculated ΔS arising from any noise or error in the measured magnetization.^[60] Decreasing the temperature step between the isotherms will increase the magnitude of the error in ΔS , leading to a trade-off between high temperature resolution and smooth data. As discussed in Section 5, the isothermal procedure has to be modified when dealing with first-order materials.

As it is often most experimentally convenient to keep the temperature constant and ramp the field, this is conventionally the route chosen. However, care must be taken when ramping the field during such experiments, even when measuring second-order materials. As the field is changed the sample temperature will change, due to the magnetocaloric effect, making the underlying assumption of a constant temperature incorrect. This is especially true close to the magnetic transition temperature, where the magnetocaloric effect is high. Recently two studies have been published illustrating the possible artifacts arising when ramping the field too fast.^[134,135]

A VSM only allows the bulk magnetization to be measured. Moore et al. have constructed a Hall probe allowing the determination of the spatial variation of the magnetization down to a resolution of a few tens of microns.^[136] The Hall probe consists of a $25 \mu\text{m} \times 25 \mu\text{m}$ InSb Hall sensor which is rastered across the surface of the sample. In this way the dynamics of the magnetic phase transitions can be studied in much more detail, allowing, e.g., first-order and second-order transitions to be clearly discriminated.^[137]

9.2. Measuring the adiabatic temperature change

The adiabatic temperature change, ΔT_{ad} , is of equal importance to the isothermal entropy change when assessing magnetocaloric materials for use in magnetic refrigeration, yet it is often not reported in the literature. Usually only measurements of the specific entropy change, ΔS , and the specific heat capacity in zero field, $C_H = 0$, are presented. This is because measuring ΔT_{ad} requires custom-built lab equipment, in which the magnetic field and the temperature can be varied independently for the magnetocaloric sample in question while the temperature is accurately measured.

The basic idea of a ΔT_{ad} measuring device is to enable the applied field to be varied in time while recording the temperature of the sample. The change in magnetic field must occur while the sample is kept quasi-adiabatic. The applied magnetic field can either be varied by keeping the sample stationary and changing

the magnetic field or moving the sample relative to a constant magnetic field. One advantage of having a stationary sample is that it makes it easier to keep the sample adiabatic, as the sample can, e.g., more easily be placed in a high vacuum container. However, the field change rate can be low, depending on the type of magnet used. If the thermal insulation of the sample is non-ideal, moving the sample relative to the magnetic field can be an advantage as a small sample can be moved quickly in and out of the magnetic field, ensuring near adiabatic conditions during the change in magnetic field. In this setup it is usually the sample that is moved, as it is always much smaller than the magnet.

In order to obtain the adiabatic temperature change at multiple values of the magnetic field, a magnetic field source where the generated flux density can be varied is needed. A common way to generate a varying magnetic field is to use an electromagnet, which is ideally suited for an experimental setup measuring ΔT_{ad} . However, an electromagnet requires substantial amount of power and generates large amounts of heat which must be kept away from the sample. Instead a permanent magnet design can be used in which the magnetic field can be varied by rotating or otherwise moving one magnet relative to another. An example of such a varying magnet design is the concentric Halbach cylinder, consisting of two nested permanent magnet cylinder rings.^[5]

The temperature of the magnetocaloric sample can be measured either directly, usually by means of a thermocouple in contact with the sample, or indirectly. An example of the latter is to measure the temperature through acoustic detection of the temperature change. Otowski et al. developed an experimental setup where a sample is placed in a small, gas tight cell and subjected to a periodically changing magnetic field.^[138] This field consists of a base field and a harmonic amplitude variation, much smaller than the base field. The frequencies of the modulation are of the order of 1–10 Hz. As the field is modulated the sample will change temperature due to the magnetocaloric effect, which gives rise to an exponentially decaying thermal wave in the gas surrounding the sample, albeit with

only a very small heat loss from the sample. The temperature of the sample is controlled by water flowing through the sample chamber. The produced thermal wave causes periodic pressure changes in the cell which can be detected with a sensitive microphone in the wall of the cell and post-processed to obtain the adiabatic temperature change. Such a thermoacoustic setup has been used to measure the adiabatic temperature change for various forms of Gd.^[138–140] Christensen et al. describe an alternative indirect approach where an IR camera is used to detect the temperature of a Gd plate which is quickly removed from the gap of a Halbach cylinder.^[141] This allows the adiabatic temperature change to be spatially resolved across the sample.

Most measurements of the adiabatic temperature change are done using direct methods. In **Figure 10** a schematic experimental setup for direct measurements is shown.^[142] The sample is made by sandwiching a thermocouple between two identical plates of the magnetocaloric material, to ensure good thermal contact. The sample is wrapped in insulating foam to reduce heat losses and placed in a sample holder such that the orientation of the sample is fixed and known relative to the sample holder. A Hall probe, which measures the flux density of the magnetic field orthogonal to the surface of the probe, is also placed inside the sample holder. The sample holder is then mounted on a pneumatic piston rod, which is used to move the sample in and out of a static magnetic field generated by a concentric Halbach cylinder design. This allows the applied field in the central bore to be adjusted between 0 and 1.5 T. The piston is automatically moved in and out of the magnetic field, while the temperature of the thermocouple is recorded. The time to move the piston in or out of the field is less than 100 ms. To measure the entire ΔT curve as a function of temperature, the sample temperature is varied by controlling the temperature of the sample environment. As the sample is not completely adiabatic it will slowly follow the temperature of the environment. As long as this temperature change is very slow compared to the movement time of the pneumatic piston, this approach can indeed be used to determine the adiabatic temperature change.

The same principle has been used to measure the adiabatic temperature change of Gd, $\text{Gd}_5\text{Si}_2\text{Ge}_2$, $\text{Gd}_5\text{Si}_{3.2}\text{Ge}_{0.8}$ and Gd_5Si_4 .^[143] Gopal et al. used a superconducting magnet and the sample chamber was evacuated to $1.3 \cdot 10^{-7}$ bar.^[144] Systems similar to this have been used to measure the adiabatic temperature change of Gd,^[145] as well as $\text{La}_{0.6}\text{Ca}_{0.4}\text{MnO}_3$,^[146] and $\text{La}_{0.67}\text{Ca}_{0.33-x}\text{Sr}_x\text{MnO}_3 \pm \delta$.^[147] Fujieda et al. measured the adiabatic temperature change of $\text{La}(\text{Fe}_x\text{Si}_{1-x})_{13}$ by quickly moving a sample into a static magnetic field.^[148]

A number of systems where the sample is kept stationary have also been presented. Benford and Brown drilled a small hole for a thermocouple in a Gd sample, which was fixed in a simple vacuum chamber.^[45] This was then placed in a superconducting solenoid, capable of generating a magnetic field of up to 7 T. The temperature was controlled by a heater, wrapped around the vacuum

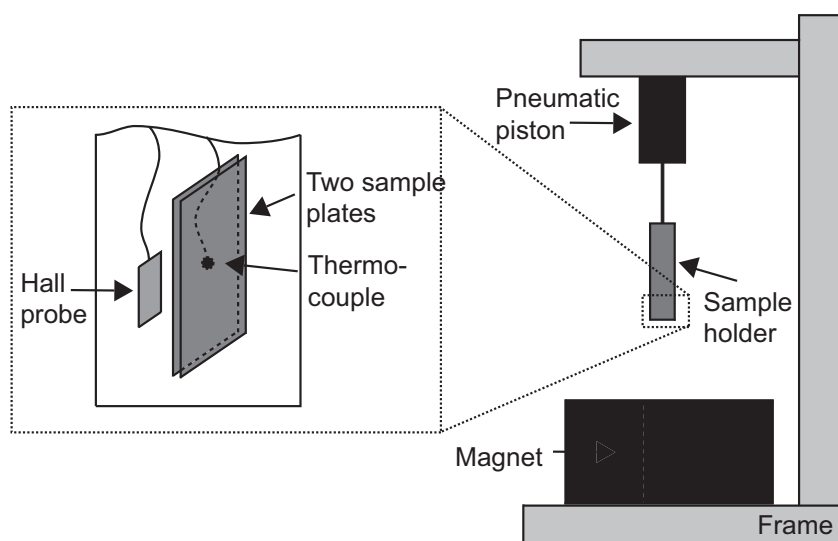


Figure 10. A schematic drawing of an experimental setup capable of measuring the adiabatic temperature change of a sample.

chamber. This is similar to the system used by Ponomarev.^[149] Dankov et al. used an electromagnet, capable of generating a magnetic field up to 2 T, to measure ΔT_{ad} .^[6] The sample was kept in vacuum at $1.3 \cdot 10^{-6}$ bar and temperature was measured using a copper-constantan thermocouple. The adiabatic temperature change was then measured by changing the field from 0 T to the desired value, which for the case of the largest field took more than 2 seconds. A detailed description of a similar device, using a high vacuum (10^{-7} bar) and a stationary sample, is given by Canepa et al.^[150] Here an applied magnetic field of up to 1 T and with a field change rate of 1 T s^{-1} was used. A similar system, used to measure the adiabatic temperature change of $\text{Mn}_{1-x}\text{As}_x\text{Sb}_x$, has also been used by Wada et al.^[151] Here a superconducting magnet was used with a field change rate of 2.5 T/min. The setup differs slightly from other devices in that during the adiabatic temperature change the temperature of the chamber is automatically adjusted to follow that of the sample, in order to reduce thermal losses.

As the conventional ramp rate of an electromagnet can be slow, a pulsed field can instead be generated by a rapid discharging of a battery. Such an approach was used by Dankov et al.^[152] producing a total pulse duration of 0.2 s with a field magnitude of up to 8 T. In order to measure the magnetocaloric response from the sample, the parasitic emf induced in the thermocouple wiring had to be subtracted from the data.

Instead of using an electromagnet, the magnetic field can also be generated by a permanent magnetic structure, as described above. Such a device is presented by Kuz'min et al.^[122] Here the magnetic field was produced by two nested Halbach cylinders with a maximum field of 1.93 T. The cylinders could be rotated quickly, producing a field change rate of 1 T s^{-1} . The temperature of the sample was measured using a thermocouple technique with an accuracy better than 0.01 K and a time resolution of 1 ms; however, the details of the experimental setup have not yet been fully presented.^[153] A similar design with a maximum magnetic field of 1.78 T and a field change rate up to 6 T s^{-1} has been demonstrated by Madireddi et al.^[154]

A number of recent publications describe various measurements of the adiabatic temperature change without discussing the experimental setup in detail. Tocado et al. measured the adiabatic temperature change of MnAs in magnetic field up to 6 T using a varying magnetic field which was changed at a rate of 0.01 T s^{-1} while the sample was kept adiabatic.^[155] Burriel et al. reported measurements of ΔT_{ad} of Mn_3GaC by keeping the sample adiabatically in a magnetic field up to 5 T generated by a solenoid.^[47] ΔT_{ad} has also been measured directly for $\text{Gd}_5\text{Si}_2\text{Ge}_{1.9}\text{Ga}_{0.1}$,^[156] as well as for $\text{Gd}_5\text{Si}_2\text{Ge}_2$.^[53] Finally, ΔT_{ad} was measured for $\text{LaFe}_{11.7}\text{Si}_{1.3}$ by use of a superconducting magnet with a field change rate of 0.7 T s^{-1} .^[157]

9.3. Specific heat measurements

There exist numerous commercial calorimeters for measuring the zero-field specific heat as a function of temperature for a given sample. However, the field dependence of the specific heat is of great importance for the characterization and performance of a magnetocaloric material. It is usually not straightforward to modify commercial calorimeters to allow for measurements in

field and therefore a number of custom-built calorimeters have been constructed for magnetocaloric characterization. It should be mentioned that measuring the specific heat as a function of field can be very time-consuming since each field value requires an entire temperature run of the calorimeter with the field kept constant. Full reports on both the temperature and field dependence of the specific heat are therefore rarely seen in the literature.

Below we discuss a number of custom-built calorimeters. Calorimeters can be broadly divided into two groups: scanning calorimeters, where the temperature is changed continuously while measuring the heat flux between the sample and a thermal bath, and heat pulse calorimeters where a small amount of heat is added at a time after which the sample temperature is measured. Using a scanning calorimeter setup with two heat flow sensors inside the same device allows the heat flux through an empty (reference) sensor to be measured at the same time as the sample. By recording the difference between the two, the precision of the measurement is greatly increased. A device utilizing this principle is called a differential scanning calorimeter (DSC). Most recent calorimeters built specifically for measuring magnetocaloric materials use the DSC approach, since they typically have lower measurement times compared to heat pulse calorimeters and do not suffer from intrinsic errors when determining the specific heat near a first-order phase transition.^[158] However, until the use of Peltier elements as heat flow sensors was introduced, heat pulse calorimeters were typically used.

An example of the latter type of device is the calorimeter constructed by Pecharsky and Gschneidner.^[159] Here a fully automated calorimeter for measuring the specific heat from 3–350 K and in magnetic field is described. This device specifically allows for an automated measurement of the specific heat, whereas previously heat pulse calorimeters required manual operation for timing of the heat pulses.

The idea of using Peltier elements as heat flow sensors is due to Plackowski et al. who demonstrated a DSC with a signal to noise ratio of 10^3 – 10^4 based on this principle.^[160] This is an inexpensive way to obtain highly reproducible data with high sensitivity. When heat passes through the Peltier element it generates a drop in voltage across the element, U , which is directly proportional to the heat flow passing through the device, $\dot{Q}$, as well as to the Seebeck coefficient, S , and inversely proportional to the thermal conductance, G , of the Peltier element:

$$U = \frac{S}{G} \dot{Q}. \quad (47)$$

Thus, the Peltier element can be used as a heat flow sensor. A sample with a size between 10 and 100 mg is typically used.^[161] The basic setup for a DSC based on this principle is shown in **Figure 11**. The signal is recorded as the difference in voltage between the sample and reference Peltier elements. By Equation 47 this directly corresponds to the heat flow through the elements. To calculate the specific heat from the differential voltage signal a calibration is performed with a sample (e.g. of Cu) of known specific heat capacity. A measurement of the voltage difference between a sample of Gd and the reference Peltier element is shown in **Figure 12**, along with the specific heat derived from it.

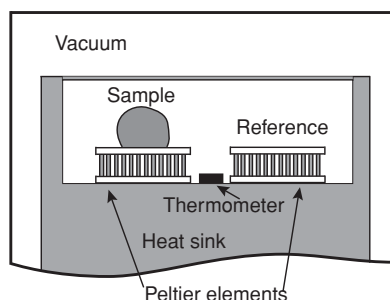


Figure 11. A schematic drawing of a differential scanning calorimeter (DSC) with two Peltier elements. Thermal contact between the sample and the Peltier element is achieved by the use of thermal grease. The lid functions as a radiation reflective shielding to minimize heat losses. The temperature inside the chamber is recorded by a PT100 thermometer and the temperature of the heat sink is controlled externally.

A number of devices similar to the original DSC using Peltier elements by Plackowski et al. have been built. Jeppesen et al. constructed a simple and high-sensitivity differential scanning calorimeter (DSC) built specifically to measure the specific heat of a magnetocaloric sample; Peltier elements are used as heat flow sensors, and the signal to noise ratio was determined to be 10^2 – 10^3 .^[161] This device has a small, relatively long cold finger which allows for sufficient room to use permanent magnets to generate the magnetic field. Also, this DSC is designed with a secondary cooling circuit, from which the temperature range can be adjusted. Marcos et al. describe a similar DSC adapted to a cryostat.^[162] The heat sensors are Peltier elements and the calorimeter is a copper spool. The temperature is controlled by the flow of a high purity helium gas with a pressure inside the calorimeter of 200–600 mbar. Convection is limited by an external copper cylinder. The typical sampling frequency is 0.25 Hz.

An entirely different concept is the thin-film nanocalorimeter constructed by Minakov et al.^[163] Here a sample of microgram size is mounted on a submicron membrane of silicon nitride

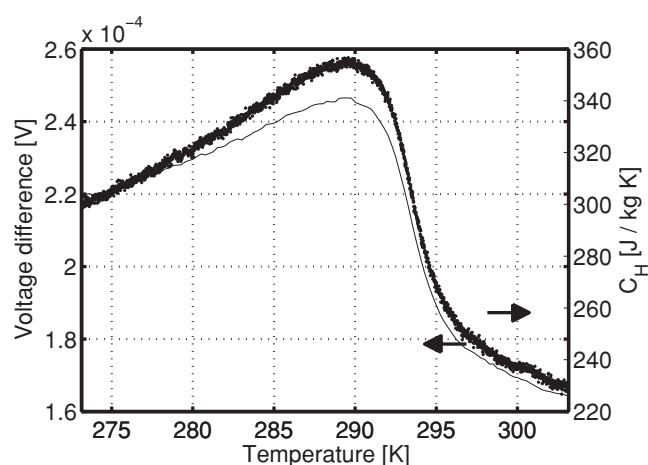


Figure 12. The voltage difference and the derived specific heat of Gd in zero magnetic field, recorded on the device described in Jeppesen et al.^[161] The high sampling frequency of the DSC data is apparent.

together with a thin-film thermopile and resistive heater. The sample is strongly thermally coupled to a thermostat through a heat exchange gas (helium at 0.1–1 bar). By passing a low frequency AC current through the heater and logging the amplitude and phase of the thermopile output it is possible to measure the heat capacity with high sensitivity. The system allows measurements up to 300 K in 8 T. An experimental procedure has been developed to allow a latent heat contribution to be distinguished from a peak in heat capacity, thus allowing unequivocal identification of a given phase transition as first or second order.^[40]

9.4. Direct measurements of the entropy change

Since the Peltier elements in the DSCs described above can be used both for measuring the heat flux and for controlling the heat flow, such calorimetric techniques can also be used to measure the isothermal entropy change directly. This is done by using the Peltier cells actively to keep the sample isothermal when the external field is changed.^[164–166] The heat required to do this is then equal to the isothermal entropy change times the sample temperature. Sasso et al. measured the history dependence in single crystals of $\text{Ni}_{55}\text{Mn}_{20}\text{Ga}_{25}$, which exhibit significant hysteresis, using this method.^[167] The same group subsequently extended the technique to compensate for the kinetic effects arising from the finite heat capacity of the measuring cell.^[168] They used it to compare the magnetocaloric effect in the second-order material $\text{La}(\text{Fe}_{1-x-y}\text{Co}_y\text{Si}_x)_{13}$ to that of the first-order, hysteretic material $\text{Ni}_{50}\text{Mn}_{36}\text{CoSn}_{13}$. The technique is particularly well-suited to first-order materials since it avoids the possible pitfalls associated with magnetization measurements.

9.5. Demagnetization effects

All of the above measurement techniques are sensitive to the effects of magnetostatic demagnetization. The magnetization of the material creates an internal demagnetizing field which tends to oppose the applied field; this field is highly dependent on the geometry and orientation of the sample with respect to the applied field. In experiments it is the applied field that can be controlled and the internal field must subsequently be determined from the measurement of the magnetization and the shape and orientation of the sample. Ideally, all experimental results should be stated in terms of the internal field to ensure that they are independent of the particular geometry of the sample used.

The internal magnetic field can in general be expressed as

$$\mathbf{H} = \mathbf{H}_{\text{appl}} + \mathbf{H}_{\text{dem}} \quad (48)$$

with $\mathbf{H}_{\text{appl}}$ and $\mathbf{H}_{\text{dem}}$ being the applied and demagnetizing fields, respectively. Only for uniform ellipsoids (as well as the infinite sheet and the infinite cylinder) is the demagnetization field constant in space.^[169–171] In that case one can write

$$\mathbf{H}_{\text{dem}} = -\mathbf{N} \cdot \mathbf{M}(T, \mathbf{H}), \quad (49)$$

where $\mathbf{N}$ is the demagnetization tensor. For an ellipsoid the demagnetizing field is determined by the principal values of $\mathbf{N}$, N_{xx} , N_{yy} and N_{zz} whose sum is unity. If both the field and the magnetization is along a principal axis, Equation 48 becomes the scalar relation

$$H = H_{\text{appl}} - N_{ii} M, \quad (50)$$

where N_{ii} is the appropriate principal value.

For other geometries, the demagnetization field and thus the internal field will vary spatially inside the sample. This means that care must be taken in choosing an appropriate shape and orientation of the sample. Since most measurements cannot resolve spatial variations of the magnetization across the sample, the usual approach to correct for demagnetization is to use a single, spatially averaged demagnetization factor N . However, when the demagnetization varies within a sample it results in different parts of the sample experiencing a range of internal magnetic fields, potentially resulting in a broadening of the measured properties. This broadening increases with the size of the demagnetization. In addition, a large demagnetization results in a significant reduction of the internal field, especially for highly magnetic materials, thus decreasing the available field range of measurement. The two points are illustrated in Figure 13. Here, the internal magnetic field of Gd, described by the mean field theory discussed above, has been modeled for a typical rod-shaped VSM sample, with the field applied along either the long axis or perpendicular to it. The applied field has in both cases been chosen so that the average internal field is 1 T. It is clearly seen how the necessary applied field to obtain an average internal field of 1 T is significantly higher for the situation with the field applied perpendicular to the long axis, and also that the distribution of internal fields is broader. Thus, although the effect of demagnetization may be corrected, it is advantageous to minimize this correction either by using

ellipsoidal samples (for which the demagnetization is spatially constant) or by measuring the sample in the minimum demagnetization orientation, i.e. along the longest axis.

Taking the demagnetization into account when measuring a powder or particle sample is also important. Even if the particles can be assumed to be spherical and thus have a constant $N = \frac{1}{3}$ they will still interact magnetically within the volume of the sample container. An approximation can be given for the average demagnetization factor in the case of a powder consisting of spherical particles with a packing ratio f ,

$$N_{\text{av}} \simeq \frac{1}{3} + f \cdot (N_{\text{cont}} - \frac{1}{3}), \quad (51)$$

where N_{av} is the average demagnetization factor of the powder sample and N_{cont} is the geometric demagnetization factor determined by the shape of the sample container.^[172]

Although demagnetization is a part of most standard textbooks on magnetism, the marked importance of demagnetization for magnetocaloric measurements was in fact only realized recently, making the interpretation of many previously published results somewhat difficult.^[173,153] As an example, for a thin plate the adiabatic temperature change may vary by 30% depending on the orientation of the sample.^[141]

10. Performance evaluation

Having now characterized a given material using the experimental methods outlined above, one may ask how the measured quantities relate to the performance of the material in a magnetic refrigeration device. As previously mentioned all three magnetocaloric quantities, ΔS , ΔT_{ad} and C_H , are of importance for the performance of a given material. A number of criteria have been suggested in the literature for a *single* number to quantify a material's potential as a refrigerative material. Wood and Potter introduced a concept they call the refrigerant capacity, RC, which they suggest should be maximized in a magnetic refrigeration cycle (and indeed in any other refrigeration cycle):^[174,175]

$$RC = \Delta S_{\text{cold}} (T_{\text{hot}} - T_{\text{cold}}), \quad (52)$$

defined for a refrigeration cycle between a cold side (temperature T_{cold}) and a hot side (T_{hot}); ΔS_{cold} is the entropy absorbed at the cold side per cycle which is then identified with the magnitude of the material's isothermal entropy change at the temperature T_{cold} . Maximizing the refrigerant capacity is then equivalent to finding the largest area rectangle which fits below a given $|\Delta S|$ vs. T curve (see Figure 14). It is important to note that in this approach the optimum hot side and cold side temperature are the result of the maximization. However, in practical applications the hot side and cold side temperature are obviously determined by the desired device specifications. Thus, a high refrigerant capacity does not imply the suitability of a material for a particular application. Two additional points must be made: 1) Wood and Potter assume that the entire magnetocaloric material is at the same temperature throughout (and thus that any regeneration is done by the heat transfer fluid, similarly to Brown's original concept described in Section 2). In

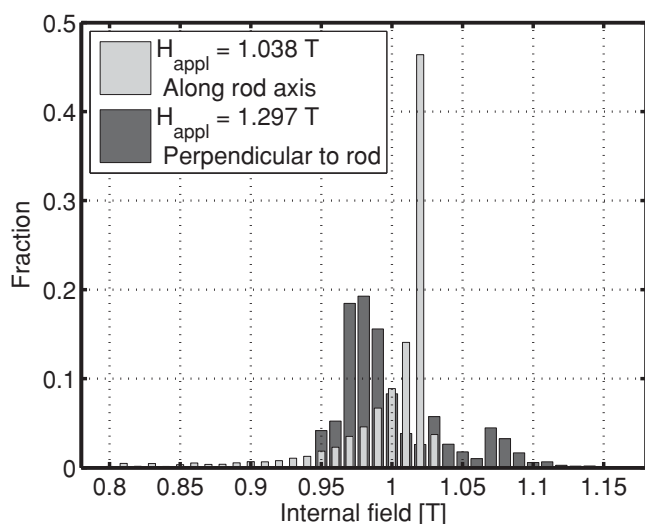


Figure 13. Distribution of the internal field in a rod-shaped VSM sample of 1 mm × 1 mm × 5 mm for two different orientations of the field. Although the applied fields are different, the average internal field, found using an averaged demagnetization factor, is in both cases 1 T.

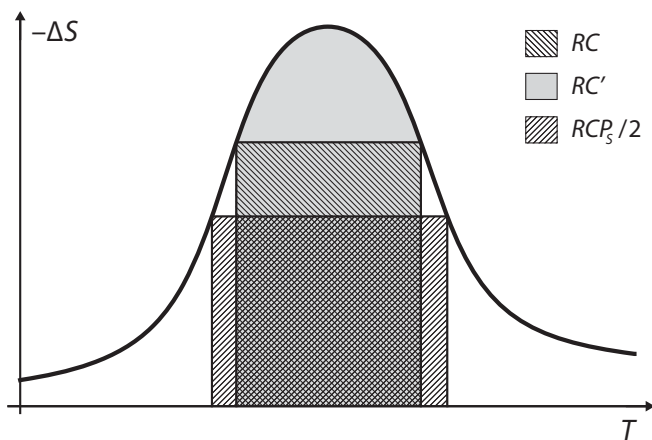


Figure 14. Three different ways to characterize the performance of a magnetocaloric material in terms of the isothermal entropy change. Shown are the areas corresponding to the refrigerant capacity RC , the integrated area (often also called the refrigerant capacity) RC' and the relative (entropic) cooling power RCP_S .

an active magnetic regenerator, there is a temperature gradient across the material, spanning from the cold to the hot side, and the regenerator as a whole does not undergo a unique thermodynamic cycle. This makes it impossible to employ the concept of refrigerant capacity directly. 2) They neglect the fact that a sufficient adiabatic temperature change ΔT_{ad} must be available at either end of the regenerator to make it possible to transfer the heat from the regenerator to the heat transfer fluid. Thus, even though the refrigerant capacity may be large for a material with a very wide $|\Delta S|$ curve, the corresponding ΔT_{ad} at the cold and the hot side may be too small.

The refrigerant capacity of Wood and Potter has a clear thermodynamic interpretation, given their assumptions. However, in the magnetic refrigeration research community another quantity is often called the refrigerant capacity, defined as the area below the ΔS curve:^[176]

$$RC' = \int_{T_{cold}}^{T_{hot}} |\Delta S(T, H)| dT. \quad (53)$$

This quantity has no obvious interpretation in terms of cycle performance; in particular it is not simply related to the amount of entropy a given regenerator is able to transport from the cold to the hot side. It also has the somewhat troubling property of being monotonously increasing for increasing temperature span $T_{hot} - T_{cold}$, giving the misleading impression that a material may be used in devices with arbitrarily large temperature spans. Even restricting the integration to, e.g., the temperature interval where $|\Delta S| > 0.5|\Delta S|_{max}$, as is sometimes done, falsely suggests that a material with a very wide but not very high $|\Delta S|$ -curve will be able to sustain a cycle with a large temperature span.

Another quantity often encountered in the literature is the relative cooling power, RCP , which is defined as the product of the maximum $|\Delta S|$ and the full width at half maximum (δT_{FWHM}) of the entropy curve. This is equivalent to the Wood-Potter refrigerant capacity (multiplied by two) with a temperature span of δT_{FWHM} and its interpretation is subject to the

same caveats. Going a step further, Gschneidner and Pecharsky suggest defining two 'relative cooling powers' corresponding to both the ΔS and ΔT_{ad} curves, respectively:^[177]

$$RCP_S = |\Delta S|_{max} \delta T_{FWHM, S} \quad (54)$$

$$RCP_T = \Delta T_{ad, max} \delta T_{FWHM, T}, \quad (55)$$

where the FWHM of the two curves in general can differ; the dimension of RCP_S is the same as that of RC (J/kg) while the dimension of RCP_T is (kelvin)². These definitions correctly recognize the need to consider both the entropy change and the temperature change when assessing the suitability of materials. (Strictly speaking it would be slightly more meaningful to evaluate the two numbers for the same temperature span, determined as the smaller of $\delta T_{FWHM, S}$ and $\delta T_{FWHM, T}$.) Unfortunately, it is the entropic relative cooling power, RCP_S , that is almost exclusively stated in the literature. Furthermore it is unclear what combination of RCP_S and RCP_T would characterize a high performance material. That the entropic relative cooling power alone is not necessarily a good indicator of performance has been shown explicitly by Engelbrecht and Bahl who show that different model materials with the same refrigerant capacity or (entropic) relative cooling power may perform very differently in a device simulation.^[178]

Of course, none of these quantities take hysteresis, if present, into account. Provenzano et al. have suggested simply subtracting the hysteretic loss, i.e. the average hysteresis area in the $M - H$ diagram for the temperature range in question, from either RC , RC' or RCP_S to arrive at an effective refrigerant capacity or entropic relative cooling power.^[179] The theoretical justification of this is not entirely clear.

It is also worth remarking that since neither RC nor RCP are linear functions of the applied field, it makes limited sense to consider, e.g., the specific refrigerant capacity, RC/H , as is sometimes done in the literature.

Usually the refrigeration capacity and the entropic relative cooling power, which both are extensive quantities proportional to the sample size, are quoted in gravimetric units, i.e. J/kg. However, Pecharsky and Gschneidner argue that from an applications point of view it is much more relevant to use volumetric (J/m³) units.^[176] Materials with low volumetric refrigeration capacity will require a higher volume of the magnet gap to achieve the same potential magnetic work per cycle, compared to materials with a higher volumetric refrigeration capacity (even though gravimetrically the latter may be worse). However, the amount of magnet material required to produce a given maximum field in a certain volume increases exponentially with the volume. Since the most significant component cost-wise of a refrigeration device is the permanent magnet,^[180] this will have a direct impact on the practical applicability of a given material.

11. Regenerator performance

Hitherto we have considered materials properties in isolation, as it were. However, in a refrigeration device the magnetocaloric material will be shaped into a certain geometry and be subjected to periodically varying magnetic fields while being in

contact with a flowing heat transfer fluid. This means that additional properties become important for evaluating the performance. These include the geometrical shape of the regenerator, the thermal conductivity of the magnetocaloric materials, and the thermal and viscous properties of the fluid. These properties will determine the size of the entropic losses associated with, e.g., the fluid pressure drop and the heat transfer processes in and between the solid and the fluid. Hysteresis, if present, will also be a source of losses. Since heat transfer between fluid and solid is necessary for the operation of the regenerator, some degree of entropic losses is unavoidable. This means that even an ideal active magnetic regenerator will never attain the theoretical Carnot efficiency of a reversible cycle. Finally, chemical and mechanical stability will be an issue for the long term operation of a regenerator.

11.1. Regenerator geometry and thermo-hydraulic properties

The geometrical shape of the regenerator is important for both the heat transfer and pressure drop properties as well as for the resulting internal magnetic field in the material. Some common options for regenerator geometries include stacked flat plates, packed spheres and wire screens. The objective is to maximize the heat transfer between solid matrix and fluid, while keeping the pressure drop and demagnetization as small as possible. The available geometries for a given choice of magnetocaloric material are to some extent constrained by processing issues. To give just one example, the cost of Gd increases at least tenfold if it has to be processed into thin plates. On the other hand, manganites and other ceramic materials may be processed into a variety of different geometries using cheap standard ceramic forming processes such as tape-casting or extrusion.^[181,182]

Efficient heat transfer between the solid magnetocaloric refrigerant and the heat transfer fluid is essential for an overall good performance of a magnetic refrigerator. It is, with the current state-of-the-art, possible to achieve the required heat transfer rates for AMR operation up to, at least, 4 Hz.^[183] However, as the (convective) heat transfer rates increase, the pressure drop across the regenerator bed tends to increase as well. A common way of evaluating the regenerator performance from a heat transfer perspective only is by considering the so-called number of transfer units, NTU. This dimensionless number is defined as the ratio between the convective heat transfer and the fluid flow rate of thermal mass in the regenerator:

$$NTU = \frac{h A_{HT}}{\dot{m}_f c_f}, \quad (56)$$

where h is the convective heat transfer coefficient, A_{HT} is the total heat transfer surface (or wetted) area, $\dot{m}_f$ is the mass flow rate of the heat transfer fluid, and c_f is the specific heat of the fluid. Generally, when NTU increases the regenerator thermal effectiveness increases.^[184]

Another important dimensionless parameter describing the performance is the ratio of the flow of thermal mass of the fluid to the thermal mass of the solid.^[4] This so-called utilization is defined as

$$\phi = \frac{\dot{m}_f c_f}{2 f m_s c_s}. \quad (57)$$

Here, f is the operating frequency of the AMR, m_s is the mass of solid regenerator material, and $c_s = C_H(T, H)$ is the specific heat of the solid. The utilization is in principle a function of temperature and magnetic field, but in the literature consensus seems to favor using a representative value of the average specific heat of the magnetocaloric material when calculating the utilization.^[76,185]

The convective heat transfer coefficient is related to the Nusselt number as

$$Nu = \frac{h d_h}{k_f}, \quad (58)$$

with d_h being the hydraulic diameter of the regenerator and k_f the thermal conductivity of the fluid. Equations 56–58 may be combined to yield

$$NTU = \frac{Nu k_f A_{HT}}{2 f m_s c_s \phi d_h}. \quad (59)$$

This expression gives an insight into the factors affecting the thermal performance of an active magnetic regenerator. The parameters that may be varied significantly are the heat transfer surface area (A_{HT}), the operating frequency (f), the thermal utilization (ϕ) and the hydraulic diameter (d_h). The Nusselt number is generally a function of the Reynolds number. For a packed particle regenerator it tends to increase as a function of the Reynolds number,^[186] whereas for geometries like flat plates or honeycomb structures the Nusselt number is virtually constant for incompressible, laminar flows.^[187]

Since the cooling power is proportional to the operating frequency, all other things being equal, it is desirable to operate at as high a frequency as possible. For the NTU to increase or at least remain constant when the operating frequency is increased it is either necessary to decrease the thermal utilization or to decrease the hydraulic diameter. This is in particular the case for flat plates and similar geometries since the Nusselt number remains constant. Engelbrecht et al. argue that when using parallel flat plate AMRs and water-based heat transfer fluids, the plate spacing should be as low as 100 μm or less to ensure high performance.^[188] This puts high demands on the tolerances of the regenerator assembly. Indeed, it has been shown that if the plates in the stack are not perfectly spaced (or very close to being so), the regenerator performance will suffer significantly.^[189,190]

A useful tool in the investigation of the purely thermo-hydraulic properties is the characterization of passive regenerators.^[184,190,191] Due to the large parameter space to be explored there still remains a lot to be done experimentally on active magnetic regenerators systematically exploring the full interplay between flow and magnetocaloric performance.

11.2. Thermal conductivity

The thermal conductivity of the magnetocaloric material making up the regenerator is important since a low value will

make the necessary heat transfer through the solid slow, while large values will increase the axial conduction (in the direction of the flow) which tends to decrease the regenerator performance. Ideally, the regenerator should have perfect conduction transverse to the direction of the flow and zero conduction in the flow (axial) direction. Of course, this ideal case cannot be attained and it is therefore important to find a compromise where the heat transfer from the solid to the fluid is as high as possible but the axial conduction at the same time is kept small.

The axial conduction may to a first approximation be considered as a time-independent loss term that is only a function of geometry of the entire regenerator bed and the temperature span across the bed. If the regenerator solid matrix is continuous along the direction of the flow (as is the case for, e.g., parallel plates) the axial conduction will in general be greater than for packed particles. However, structures like parallel plates have a significantly reduced pressure drop compared to packed particles (whether irregular granulates or spheres) and the regenerator bed can therefore be made longer, which will reduce the axial conduction significantly. High frequency operation will also decrease the significance of the axial conduction due to the increased number of active thermal regenerations per time.

An interesting idea is to tailor the thermal conductivity by impregnating, e.g., small amounts of silver into porous magnetites and in this way increase the thermal conductivity of the magnetocaloric material.^[192] This will of course degrade the magnetocaloric performance compared to a fully dense sample, as one is replacing magnetocaloric material with non-magnetocaloric material. On the other hand device performance may improve. There have as yet been few studies attempting to identify the optimal balance between magnetocaloric effect and thermal conductivity.^[193]

11.3. Demagnetization effects

Demagnetization not only plays a role for materials characterization. It is also crucial to consider it in the context of regenerator performance. The demagnetization will depend on temperature due to the temperature dependence of the intrinsic magnetization. Thus, the effect of demagnetization will not only be to reduce the internal field but also to introduce a position and temperature dependence of the field inside the regenerator. In Figure 15 the average internal field in a rectangular prism is found as a function of the aspect ratio of the prism (keeping two dimensions constant and varying the third). The results clearly show that the internal field is a function of both temperature and geometry of the magnetized object. The impact of the demagnetizing field on the AMR performance is not understood in full detail yet. However, purely magnetic models have been developed that are able to calculate the internal field in various structures.^[194–196] It is generally found that the internal magnetic field is typically around 10–20% lower than the applied field but it can be reduced by as much as 80% in certain cases. It is therefore a significant effect that should be considered carefully when modeling and designing a magnetic refrigerator.

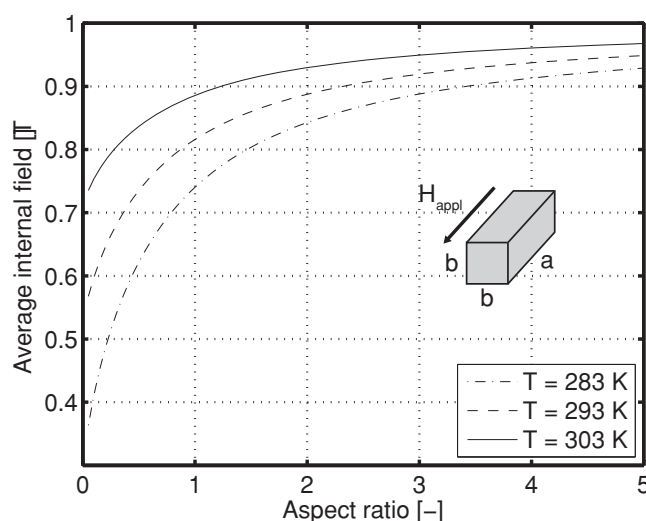


Figure 15. Modelling results for the average internal field in a rectangular prism with dimensions $a \times b \times b$ with varying aspect ratio a/b . The applied field of 1 T is in the direction of a and the assumed material properties are those of mean field gadolinium with a Curie temperature of 293 K.

11.4. Hysteresis

First-order materials can generally have a higher magnetocaloric effect than second-order materials since they can harness part of the entropy associated with the lattice degrees of freedom. On the other hand, hysteresis can be the source of irreversible losses. Therefore it is a matter of some interest to understand the impact of hysteresis on AMR performance. However, relatively little work has been done in this field to date. Basso et al. have tried to apply non-equilibrium thermodynamics based on a Preisach model (cf. Section 6.6) to both Carnot and AMR cycles.^[100] They show that the irreversibility of the material will act as a source of losses. However, their AMR cycle analysis is too simplified to allow general conclusions to be drawn. To the knowledge of the present authors no satisfactory hysteresis model has been implemented in a detailed AMR model. Recent studies of the directly measured adiabatic temperature change in $\text{LaFe}_{11.8}\text{Si}_{1.2}$ and $\text{Ni}_{2.21}\text{Mn}_{0.77}\text{Ga}_{1.02}$ samples that exhibited hysteresis show that the reversible ΔT_{ad} that is most likely experienced by the regenerator in an AMR device is very close to the adiabatic temperature change calculated from the low-field, heating entropy curve and the high-field, cooling entropy curve;^[197] similar conclusions have been reached from the study of $\text{La}(\text{Fe},\text{Si})_{13}\text{H}_y$.^[198] In other words, the material in an actual device experiences the full hysteresis. Therefore, it seems that the material properties of an MCM that exhibits hysteresis can be estimated by assuming the properties at zero field to be those measured as the material is heated and properties at higher fields are those measured as the material is cooled. This will reduce the magnetocaloric effect in the material proportionally to the hysteresis.

11.5. Chemical and mechanical stability

The magnetocaloric material in a regenerator is in contact with a flowing heat transfer fluid. This means that the corrosion

stability of the material has to be considered. Several of the metallic magnetocaloric materials are known to corrode in water but the question has not been studied extensively in the open literature. Zhang et al. have studied the corrosion resistance of Gd in water and a NaOH solution,^[199,200] while Chen et al. found some passivation of Gd in an oxalic acid solution.^[201] Hansen and Nielsen tested the corrosion resistance of $\text{La}(\text{Fe},\text{Co})_{11.9}\text{Si}_{1.1}$ in a number of commercially available fluids.^[202] Engelbrecht et al. found that standard automotive antifreeze offers satisfactory corrosion protection when using Gd in a device.^[203] Sintered ceramic materials are expected to exhibit a good corrosion resistance at room temperature but no systematic studies have been published. Proprietary corrosion inhibitors are used by some device developers.^[204]

Another issue to be considered when magnetocaloric materials are used in an AMR is the mechanical stability. The periodic field changes will induce magnetic forces which the regenerator matrix must be able to withstand. Furthermore, when using first-order materials there may be a significant volume change associated with the phase transition which may induce cracks and other mechanical defects.^[205] Again, this is a topic which has not been explored sufficiently.

12. Graded regenerators

In steady state operation there will be an overall temperature gradient inside an active magnetic regenerator in the direction from the cold side to the hot side. Each cross-section of the regenerator will only experience small temperature deviations from the local average temperature. Since the magnetocaloric effect of a given material peaks at a single temperature close to the Curie temperature and falls off rather quickly away from the peak temperature, it follows that only a single cross-section of a one-material regenerator will actually experience the maximum magnetocaloric effect. This leads to the concept of graded regenerators, i.e. regenerators whose composition varies spatially along the direction from the cold side to the hot side such that the Curie temperature at a given cross-section is close to the local average temperature during operation. Ideally, the regenerator would be continuously graded. In practice, regenerators consisting of two to five materials have been tested (see Section 14, below).

Some authors have considered multi-phase materials where compositions with different Curie temperatures are mixed.^[206,207] Although such mixtures may have a larger refrigerant capacity (as defined by Equation 53) than each constituent material alone, when used in a regenerator only a fraction of the material will be performing optimally at a given point. Thus, mixtures will in general have a poorer performance than a graded regenerator made of the same materials. This is a further illustration of the point that the refrigerant capacity in itself is of little predictive value for regenerator performance.

Both modeling and experiments suggest that a carefully graded regenerator will significantly increase the performance of the regenerator in a refrigeration device.^[208] There are, however, several very interesting unanswered questions in this field: what is the optimal grading of a specific regenerator? How is the grading to be implemented in an optimal and cost-effective

way in an actual device, and how are the different compositions of the magnetocaloric materials chosen to ensure that they are compatible? What is the resulting internal magnetic field in the regenerator? These topics are not fully understood yet, although a few preliminary modeling results have been reported.^[209]

13. Materials

It is not the intent of the present section to give a comprehensive review of magnetocaloric materials. As mentioned in the introduction, the rapidly growing literature is well covered by a number of recent reviews. Here the emphasis will be on a number of important materials groups which show promise for actual application and with an emphasis on materials which can be used for graded regenerators.

13.1. Gd and its alloys

As the only pure element with a Curie temperature near room temperature gadolinium has been extensively investigated. In many ways it can be considered as the best performing material available: the adiabatic temperature change is large, the specific heat fairly low (around 300 J/kg·K at room temperature) and the thermal conductivity high. The detailed properties of gadolinium depend rather strongly on the impurity content.^[6] Thus, care has to be taken when comparing results from different research groups and indeed when modeling and evaluating Gd-based AMR systems.

A variety of alloys between gadolinium and another lanthanide metal (e.g. Tb, Dy, Ho, and Er) have been investigated for their magnetocaloric properties.^[2] The general effect is to lower the transition temperature compared to Gd while maintaining a comparable magnetocaloric effect. This makes them attractive for graded regenerators. The main drawback of Gd-based regenerators is probably cost, both materials cost and shaping cost.

13.2. $\text{Gd}_5\text{Ge}_2\text{Si}_2$ and similar compounds

$\text{Gd}_5\text{Ge}_2\text{Si}_2$ was the first 'giant' magnetocaloric material to have been discovered and its detailed properties have been investigated by several authors, as discussed by Gschneidner et al. in a recent review.^[8] It is one member of the family $\text{Gd}_5(\text{Si}_{4-x}\text{Ge}_x)$ which show very interesting magnetocaloric and other properties.^[210] Indeed their magnetocaloric properties exceed those of Gd. Increasing x from 0 to 2 allows one to change the Curie temperature from 336 K to around 300 K. At $x = 2$ the crystal structure changes from orthorhombic to monoclinic. It is possible to reduce hysteresis significantly by the addition of small amounts of Fe or certain other metals.^[179,156]

13.3. $\text{La}(\text{Fe},\text{Si})_{13}$ and similar compounds

A number of intermetallic compounds with the cubic NaZn_{13} structure exhibit a sizable magnetocaloric effect. The hypothetical intermetallic compound LaFe_{13} does not exist, but it

turns out that with the substitution of above 10% Si or Al for Fe the structure may be stabilized. Hu et al. in 2001 found a large magnetocaloric effect in $\text{LaFe}_{11.4}\text{Si}_{1.6}$ around its first-order transition temperature of 208 K.^[211] By introducing interstitial hydrogen in the structure T_c may be tuned over a large temperature interval: the transition temperature of $\text{La}(\text{Fe}_{0.88}\text{Si}_{0.12})_{13}\text{H}_y$ changes from ~200 K to ~330 K as y increases from 0 to 1.5.^[212] Another way of changing the Curie temperature is by partially substituting Fe with, e.g., Al, Co or Mn, while La may be partially substituted with Pr; substitution may change the transition from first to second order.^[213,214] Shen et al. have recently reviewed a substantial number of these compounds.^[215] When investigating such compounds it is important to consider long term stability. Barcza et al. found that partially hydrogenated ($y < 1.5$) samples decompose into two different phases over several days; instead fully hydrogenated Mn-containing compounds were found to be stable while still allowing the Curie temperature to be adjusted.^[216]

The materials cost for these materials is comparatively low, and a powder metallurgical production method suitable for large scale production has been developed.^[217] Materials produced by this route show low hysteresis, making them suitable for device operation.^[218] Alternatively, hysteresis and mechanical degrading may be controlled by introducing porosity in the structure, as demonstrated by Lyubina et al.^[205,219]

13.4. MnAs and similar compounds

Intermetallic compounds containing Mn is another large class of interesting magnetocaloric materials. They have been comprehensively reviewed by Brück et al.^[220] MnAs exhibit a hysteretic first-order magnetostructural transition around 317 K with an isothermal entropy change of the order of 40 J/kg·K for a field change of 0 → 2 T, and only increasing very slowly with increasing field.^[46,221] The transition temperature and hysteresis behavior may be adjusted in a variety of ways by substitution. Substituting Sb for As both lowers the transition temperature and reduces hysteresis significantly,^[222] while $\text{MnFeP}_{0.45}\text{As}_{0.55}$ was found to have a transition temperature in the vicinity of 300 K and low hysteresis.^[223] By adjusting the P/As ratio the transition temperature can be varied from 150 to 335 K.^[224]

The $\text{MnFeP}_{1-x}\text{As}_x$ materials generally show great promise for magnetic refrigeration. One challenge is that they contain arsenic and one has to be certain that no As is released to the surroundings during operation in a device; so far the toxicity of these compounds has not been extensively studied.^[220] This question also has to be considered in conjunction with the small volume change suffered by the compounds during the first-order transition which may lead to mechanical issues.

There has been a considerable research effort directed towards the development of compounds without As. Promising systems include $\text{MnFe}(\text{P},\text{Si},\text{Ge})$ and $(\text{Mn},\text{Fe})\text{PSi}$. Cam Thanh et al. found a large magnetocaloric effect in $\text{MnFe}(\text{P}_{0.89-x}\text{Si}_x)\text{Ge}_{0.11}$ and similar compounds, with a tunable transition temperature in the range 250–300 K.^[62,225] In $(\text{Mn},\text{Fe})_{1.95}\text{P}_{0.50}\text{Si}_{0.50}$ Dung et al. found that the hysteresis may be tuned by varying the Mn/Fe ratio and that the order of the transition may be changed from first to second order.^[226]

13.5. Manganites

The perovskite manganites are a fascinating class of materials with many intriguing magnetic properties, including colossal magnetoresistance and multiferroic phases. They have also been widely investigated for their magnetocaloric properties.^[227] They can be described by the general formula $\text{R}_{1-x}\text{A}_x\text{MnO}_3$ where R is a trivalent rare earth ion (e.g. La, Pr, Nd) and A is a divalent alkaline earth element (e.g. Mg, Ca, Sr, Ba). For a given choice of elements it is possible to vary the Curie temperature significantly by changing the composition. Thus, the series $\text{La}_{0.67}\text{Ca}_{0.33-x}\text{Sr}_x\text{MnO}_3$ gives Curie temperatures in the range 267–370 K with an approximately linear relation between T_c and x when x is varied between 0 and 0.33.^[147] For these materials the maximum adiabatic temperature change is about 1–1.5 K at an applied field of 1 T, while $(-\Delta S)_{\text{max}}$ is 2–3 J/kg·K, i.e. somewhat lower than Gd. This is to be expected for second-order manganites: since the magnetic order is carried only by the Mn atoms the magnetocaloric effect is diluted. However, lower price, corrosion resistance, easy tunability of the Curie temperature, and good processability may still make them relevant for device application.

14. Device performance

A considerable number of magnetocaloric devices based on the active magnetic regenerator principle have been described in the literature. All such devices include a regenerator, a magnet array, a flow circuit (including a pump or a displacer) for the heat transfer fluid, some form of hot side and cold side heat exchangers, and a source of mechanical power (e.g. a motor). This makes the design space rather large. At the present stage of device development experience shows that the performance of a given device depends very much on the entire design and not only on the regenerator performance. This makes it difficult to compare the detailed performance of different materials if they are not tested in the same device and with comparable regenerator geometry. This has only been done to a limited extent so far.

From a device point of view the overall performance may be characterized by a small number of parameters: the temperature span, $\Delta T_{\text{span}} = T_{\text{hot}} - T_{\text{cold}}$, the cooling power, $\dot{Q}$, i.e. the amount of heat transferred per unit of time from the cold side to the hot side, and the coefficient of performance, COP, defined as the ratio of the cooling power to the power $\dot{W}$ consumed by the device:

$$\text{COP} = \frac{\dot{Q}}{\dot{W}}. \quad (60)$$

The three quantities ΔT_{span} , $\dot{Q}$, and COP are not independent. For given operation conditions (flow rate, operation frequency, etc.) $\dot{Q}$ will be a function of ΔT_{span} , mapping out a so-called cooling curve which has a maximum, $\dot{Q}_{\text{max}}$, for $\Delta T_{\text{span}} = 0$ and will go to zero in $\Delta T_{\text{span, max}}$ (see Figure 16). It is important to note that neither of the extreme values of the curve are of interest for actual operation: $\dot{Q}_{\text{max}}$ is only achieved with no temperature span, while the device operating at a temperature

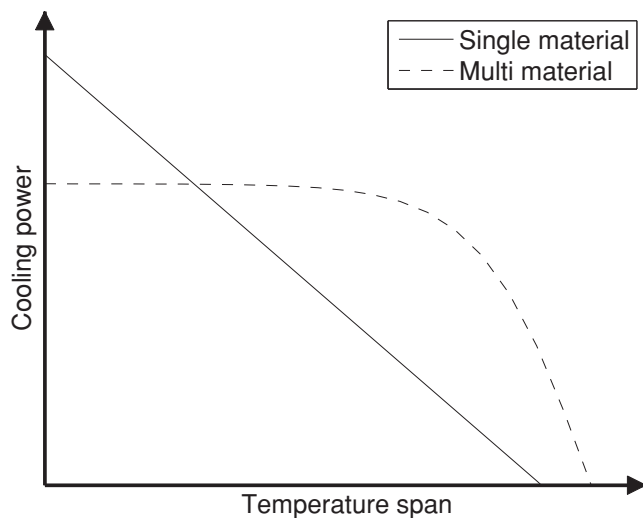


Figure 16. Schematic cooling curves for a refrigeration device containing respectively a single-material regenerator and a graded regenerator.

span of $\Delta T_{\text{span, max}}$ will not be able to sustain a heat load. Even so, many device reports only quote the two extremal values. To make it possible to compare different devices and to identify the critical parameters for performance it is necessary to map out complete cooling curves for at least a representative set of different operation parameters.

Of ultimate interest is not only that a given device will be able to operate at a relevant temperature span with an appropriate heat load, but also that it will be able to do so with a high efficiency, as measured by COP. The theoretical limit to COP is given by the Carnot value:

$$\text{COP} < \text{COP}_{\text{Carnot}} = \frac{T_{\text{cold}}}{\Delta T_{\text{span}}} \quad (61)$$

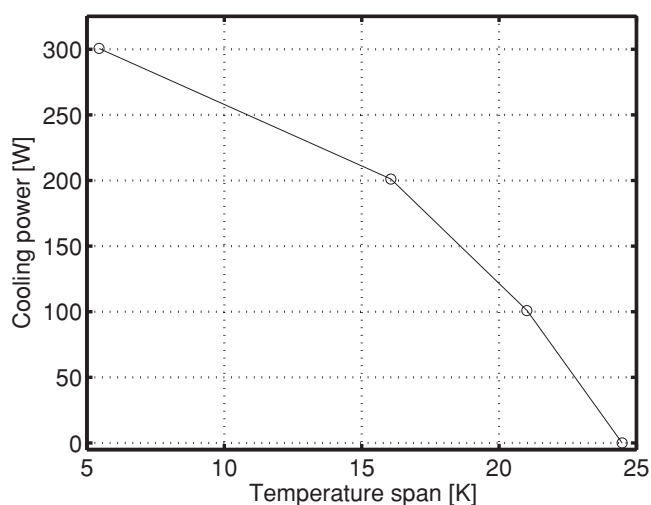


Figure 17. Cooling curve for a magnetic refrigeration device designed and built at the Technical University of Denmark. The device contains 2.8 kg Gd in the form of 0.25–0.8 mm spheres and can be operated at cycle frequencies up to 10 Hz.^[181,231]

When operating the device at the maximum temperature span, COP will be 0. At the other extreme, COP can in theory increase without limit as the temperature span approaches zero. Again, the relevant value will be the COP at the actual operation conditions. Modeling studies have indicated that magnetocaloric devices may achieve efficiencies well above those of traditional vapor compression units.^[228] However, COP is notoriously hard to model satisfactorily since it depends critically on the entire device design, including heat leaks and friction in moving parts. The usual approach is simply to take the calculated pressure drop through a given regenerator, converting it to a pump power, and adding the magnetic work input as calculated from the model. This will certainly overestimate the COP of a given device. On the experimental side, COP is not often quoted in device studies, although it is in principle not hard to measure.

Most of the devices reported in the literature use Gd, with two regenerator geometries typically considered: spheres or flat plates.^[38,183,235,236] Performance differs widely, as shown in Table 1. A lab device designed and constructed at the Technical University of Denmark recently achieved a 100 W cooling load at a temperature span of 20.5 K, using a regenerator with packed Gd-spheres (Figure 17).^[231] Several devices have been tested with Gd in combination with its alloys, e.g. $\text{Gd}_{1-x}\text{Tb}_x$ and $\text{Gd}_{1-x}\text{Er}_x$, in a graded (layered) regenerator configuration, showing that they can outperform single material regenerators.^[232,237,238] Arnold et al. find that the largest temperature span is obtained when each material during operation has an average temperature close to its Curie temperature.^[185]

Yu et al. used a $\text{Gd}_5\text{Si}_2\text{Ge}_2$ regenerator in the form of crushed particles of sizes 0.3–0.75 mm in a reciprocating device which produced 10 W of cooling over a 3 K span.^[229] The same device running with a Gd regenerator produced 19 W over a 4 K temperature span. The effect of doping the $\text{Gd}_5\text{Si}_2\text{Ge}_2$ with Ga was also studied: adding Ga was shown to increase the no-load temperature span of the device from a maximum of 10 K for $\text{Gd}_5\text{Si}_2\text{Ge}_2$ to 25 K for $\text{Gd}_5\text{Si}_{2-x}\text{Ge}_{2-y}\text{Ga}_{x+y}$.^[233]

A five-material regenerator was constructed by Russek et al. at Astronautics Corporation using crushed $\text{La}(\text{Fe},\text{Si})_{13}\text{H}_y$ particles. It produced a cooling power of 300 W at a 13.5 K temperature span while the same device operating at the same conditions produced 150 W of cooling at a 13.5 K span using a Gd regenerator.^[204] However, the no-load temperature span was higher for the Gd regenerator than for the $\text{La}(\text{Fe},\text{Si})_{13}\text{H}_y$ regenerator. The performance of $\text{La}(\text{Fe},\text{Co},\text{Si})_{13}$, $\text{La}_{0.67}\text{Ca}_{0.33-x}\text{Sr}_x\text{MnO}_3$ and Gd flat plate regenerators has been compared on the same device by Engelbrecht et al.^[203] A single-material $\text{La}(\text{Fe},\text{Co},\text{Si})_{13}$ regenerator achieved a maximum no-load temperature span of 6.9 K, compared to an 8.8 K temperature span for Gd under similar operating conditions. Using a two-material $\text{La}(\text{Fe},\text{Co},\text{Si})_{13}$ regenerator increased the no-load temperature span to 8.5 K, while a single-material $\text{La}_{0.67}\text{Ca}_{0.33-x}\text{Sr}_x\text{MnO}_3$ regenerator produced a temperature span of 5.1 K (about five times the maximum adiabatic temperature change in the given field). A two-material regenerator using graded, tapecast platelets with nominal composition $\text{La}_{0.67}\text{Ca}_{0.2925}\text{Sr}_{0.0375}\text{Mn}_{1.05}\text{O}_3$ and $\text{La}_{0.67}\text{Ca}_{0.2850}\text{Sr}_{0.0450}\text{Mn}_{1.05}\text{O}_3$ produced a temperature span above 9 K in the same device.^[234]

Although steadily improving, device performance is still some distance away from the performance demanded by

Table 1. Summary of the performance of some recent laboratory devices discussed in the text.

Magnetocaloric material	Mass (kg)	Regenerator geometry	$\mu_0 H_{\max}$ (T)	ΔT_{span} (K)	$\dot{Q}_c$ (W)	Reference
Gd	0.93	spheres (0.3 mm)	1.4	4	19	Yu et al. ^[229]
Gd	0.89	spheres (0.25–0.35 mm)	1.4	13.5	150	Russek et al. ^[230]
Gd	0.11	spheres (0.3 mm)	1.5	29	0	Tura and Rowe ^[183]
				10	50	Tura and Rowe ^[183]
Gd	2.8	spheres (0.2–0.8 mm)	1.2	20.5	100	Engelbrecht et al. ^[231]
				25.4	0	Engelbrecht et al. ^[231]
Gd	0.08	plates (0.9 mm; spacing 0.6 mm)	1.0	8.8	0	Engelbrecht et al. ^[203]
Gd/Gd _{1-x} Er _x /Gd _{1-x} Tb _x	0.12	particles (0.55 mm)	2.0	50	0	Rowe and Tura ^[232]
				25	10	Rowe and Tura ^[232]
Gd ₅ Si ₂ Ge ₂	1.21	particles (0.3–0.75 mm)	1.4	3	10	Lu et al. ^[233]
				10	0	Lu et al. ^[233]
Gd ₅ Si _{2-x} Ge _{2-y} Ga _{x+y}	1.2	particles (0.3–0.75 mm)	1.4	25	0	Lu et al. ^[233]
La(Fe,Si) ₁₃ H _y (5 materials)	0.7	particles (0.25–0.4 mm)	1.4	13.5	300	Russek et al. ^[230]
La(Fe,Co,Si) ₁₃	0.07	plates (0.9 mm; spacing 0.6 mm)	1.0	6.9	0	Engelbrecht et al. ^[203]
La(Fe,Co,Si) ₁₃ (2 materials)	0.07	plates (0.9 mm; spacing 0.6 mm)	1.0	8.5	0	Engelbrecht et al. ^[203]
La _{0.67} Ca _{0.33-x} Sr _x MnO ₃	0.03	plates (0.3 mm; spacing 0.2 mm)	1.0	5.1	0	Engelbrecht et al. ^[203]
La _{0.67} Ca _{0.33-x} Sr _x MnO ₃ (2 materials)	0.05	plates (0.3 mm; spacing 0.4 mm)	1.0	9.3	0	Bahl et al. ^[234]

real-world applications. There is a significant amount of work to be done on the device design and construction side. But probably just as important is the need for detailed performance characterization of existing devices combined with systems and regenerator modeling. Only in this way will it be possible to gain a better understanding of the complex interplay between materials properties, regenerator geometry and device design which determines the final performance.

15. Outlook

Research in magnetocaloric materials and their application in refrigeration devices and heat pumps has been a rapidly growing field for the past 15 years, if not longer. Significant progress has been made in many areas, including discovery of new groups of materials with a large magnetocaloric effect, improved understanding of many aspects of both the microscopics and the thermodynamics of the materials, and an increasing level of sophistication in the regenerator and device modeling and design. In this concluding outlook we will briefly point to some interesting and relevant directions for future research.

A large number of materials have been synthesized and summarized by magnetization measurements. However, if such materials are to be assessed for performance there is a need for a much more detailed characterization, including the field-dependent specific heat capacity and direct measurements of the adiabatic temperature change. It is also important to bear in mind that if a given material is to find application in a device, some form of shaping process must be applied to it. The interrelation between microstructure, shaping and performance has to be explored in much more detail for many of

the proposed materials in the literature. This is also of great relevance to the problem of how to achieve an optimal grading of the regenerator properties; as we have seen above there are still many open questions in this connection.

Several new materials concepts also seem worth exploring in greater detail. These include possible enhancements of the magnetocaloric effect in nanomaterials or nanocomposites,^[239–242] in thin films,^[243–245] and in superlattices/heterostructures.^[264–251] Whether such concepts will also be relevant on the scale of an actual device remains to be seen.

For first-order materials some important progress has been made in the understanding of the thermodynamics of the hysteresis and the metastable mixed phases associated with it. However, much more can be done, in particular in improved characterization of the mixed state, resolved both spatially and temporally, and how its properties relate to the hysteresis observed. A question of particular relevance for application of first-order materials is the extent to which hysteresis is an intrinsic property of some materials or whether it always is a consequence of a particular manufacturing route giving rise to a specific microstructure. On the theoretical side, it would be interesting to develop a more unified approach to hysteresis by integrating the generic hysteresis description with microscopically derived model Hamiltonians.

The impact of hysteresis on actual device performance remains largely unexplored. This is certainly a question which needs to be addressed carefully if the full potential of many of the proposed magnetocaloric materials is to be realized. This should be done both by modifying the existing active magnetic regenerator models to be able to include hysteresis, but also by systematic performance studies on hysteretic materials in actual devices. Indeed, this is true not only for hysteretic materials:

Only relatively few materials have been tested in a device. And even when a material has been tested in one device, a comparison of results obtained with a different material in another device is not easy to correlate directly to the difference in materials properties. This means that systematic studies of different materials should be undertaken using the same regenerator geometry in the same device. Such studies would be extremely valuable not only for improving regenerator and device designs but also for giving feedback to the materials development.

Other physical effects closely related to the magnetocaloric effect may also be of interest for refrigeration purposes. Many of the materials mentioned in this review also exhibit a barocaloric effect, i.e. a reversible temperature change in response to an external pressure change. This is of course due to the similar impact on the lattice entropy of a magnetic field and an external pressure near a magneto-structural phase transition. Several authors have studied the effect and proposed its utilization in refrigeration devices.^[252–256] Also electrocaloric materials where an electric field can be used to align the polarization and thus give rise to a temperature change are of interest, as summarized in a recent review.^[257] However, the latter materials are largely unrelated to magnetocaloric materials.

On the device side, development has so far focused on the active magnetic regenerator concept. There is still room for considerable improvement in the design of such devices. A number of new device concepts have also been proposed, including having solid state heat transfer using heat switches.^[258] Much remains to be done in this area, both as regards modeling and device design. For now, the most promising direction for continued device development remains the AMR concept. Even at the present state of the science and technology of room-temperature magnetocalorics we are at the brink of seeing devices attaining a commercially relevant performance (although not yet at a commercially relevant price). This bodes well for the future of this fascinating cross-disciplinary field which brings together solid state physics, materials science, heat transfer modeling and device engineering.

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